

Help Converts Newcomers, Not Veterans: Generalized Reciprocity and Platform Engagement on Stack Overflow

Abstract

Whether people who receive help on an online platform go on to help others is an open empirical question. Prior observational studies report mixed results, and designs that do not account for how active a user already was may overstate reciprocity. We develop a matched difference-in-differences survival analysis of over 21 million Stack Overflow questions, comparing users whose questions received an answer with propensity-score-matched users whose questions did not. Users whose questions were answered went on to answer other people’s questions slightly more often than their matched counterparts (increment at answer arrival, hazard ratio $[HR] = 1.049$, 95% CI $[1.044, 1.054]$). How strongly users responded depended on how long they had been on the platform: newcomers responded most ($HR = 1.07$ in the first week), while the longest-tenured users did not respond measurably at all. This newcomer pattern is the most robust in the data, and it is present in the pre-2020 period as well, before generative-AI tools arrived. How quickly help arrived also mattered: users answered within 30 to 60 minutes responded most strongly, while the very fastest answers, arriving in under 30 minutes, show no elevation at all. We read this as qualified support for an immediacy account, though we cannot fully separate how quickly an answer arrived from how good it was. Our data identify short-run behavioral patterns, not the psychological mechanisms (gratitude, moral obligation, incentive displacement) that theory proposes. The pattern is consistent with generalized reciprocity working less as a general engine of cooperation than as a contributor-recruitment channel at the point of entry, before platform-specific incentives take hold: designers who want more people answering questions gain most by making sure newcomers get answers.

Keywords: generalized reciprocity, online communities, knowledge sharing, survival analysis, platform design, Stack Overflow

1. Introduction

Whether people who receive help on an online platform go on to help others has been a persistent question in the study of online communities. Generalized reciprocity, the tendency to help others after being helped oneself (Nowak and Roch, 2007; Baker and Bulkley, 2014), is among the mechanisms proposed to sustain cooperation on platforms like Stack Overflow and Wikipedia, where the same two people rarely trade help directly. If people who receive help reliably go on to give it, cooperation may sustain itself through cascading helping acts, even among strangers (Fowler and Christakis, 2010). The behavioral evidence, however, is inconsistent: some observational studies find substantial reciprocity effects (Baker and Bulkley, 2014; Chen et al., 2019), while others find weak or null relationships (Wasko and Faraj, 2005; Joyce and Kraut, 2006). We argue that much of this inconsistency is methodological. Our contribution is a cleaner short-run test on one platform: whether users help more after someone answers their own question. The data do not directly measure the psychological mechanisms that theory proposes (gratitude, moral obligation, incentive displacement); we test only the behavioral consequence those mechanisms predict.

This question has acquired new urgency. As generative-AI assistants increasingly answer the questions people once brought to human communities, the human-to-human helping our design measures is thinning, and the reciprocal exchange that may recruit new contributors could thin with it. Our sample runs through early 2025 and so predates the steepest of that shift, which makes it a clean pre-substitution baseline: it quantifies what a human answering ecosystem did for contributor recruitment while that ecosystem was still intact — a measurement that may not be possible to make again.

Researchers who study reciprocity on platforms face a persistent confound: the most engaged users both seek help and give it, whether or not their own questions get resolved. Studies that correlate help received with help given, without accounting for how active a user already was, conflate reciprocity with general engagement (Baker and Bulkley, 2014; Chen et al., 2018; Joyce and Kraut, 2006). Prior designs have also largely treated receiving help as a one-time event, even though people ask for and give help continuously.

We address both problems by developing a matched difference-in-differences

(DiD) survival analysis, applied to over 21 million questions from Stack Overflow. For each question a user posts, we define observation windows before and after the question and compare changes in helping behavior between users whose question received an answer (treated) and propensity-score-matched users whose question did not (control). We use Cox proportional hazards models, which accommodate both the continuous timing of helping and the fact that answers arrive at different moments for different users.

We develop two theoretical predictions alongside the basic reciprocity hypothesis (H1). Drawing on theories of socialization and norm internalization, we predict that the reciprocity effect is stronger among newcomers than among experienced users (H2). Newcomers have not yet internalized community-specific norms or learned to value platform incentives such as reputation points; general moral obligations to reciprocate are therefore more likely to drive their behavior. As users accumulate experience, these general-purpose triggers may be displaced by community-specific motivations. We also predict that faster responses amplify the reciprocity effect (H3), because gratitude is more intense when help arrives while the need is still salient (McCullough et al., 2001; Tsang, 2006).

The results support the baseline reciprocity hypothesis (H1) in a modest, newcomer-concentrated form, and offer qualified support for the tenure gradient (H2) and the immediacy pattern (H3). Compared with their matched controls and with their own pre-question baseline, users who receive an answer go on to answer other people’s questions slightly more often: the pooled increment at answer arrival is $HR = 1.05$ (95% CI [1.04, 1.05], $p < .001$). How strongly users respond depends on how long they have been on the platform: newcomers respond most ($HR = 1.07$ in the first week, $p < .001$), and the longest-tenured users do not respond measurably at all. Throughout, we estimate the increment at answer arrival — the difference-in-differences contrast that removes any treated–control divergence arising before the answer. We report the waiting-period term separately, as a parallel-trends check rather than part of the effect, and we re-estimate the model under broader outcome definitions that add lower-cost helping acts (comments, edits) to show where the signal resides (Table 6).

Our study makes three contributions. First, using a design built to remove the activity confound that inflates observational estimates, we provide a cleaner short-run test: users who receive an answer help slightly more afterward (arrival-increment $HR = 1.05$), and by much less than prior observational studies report. Second, newcomers account for nearly all of that

response and veterans show none — the tenure gradient is the most robust and informative pattern in the data, though we cannot cleanly separate reciprocity from selection into being answered. Third, methodologically, we decompose the outcome by help type and enter the waiting period as an explicit term, making pre-answer divergence visible rather than silently absorbed into the treatment effect.

2. Theory

2.1. *Generalized Reciprocity*

A central mechanism proposed to support cooperation in online communities is *generalized reciprocity*, the tendency for individuals who have received help to help others in turn, even when the recipient is not the original benefactor (Nowak and Roch, 2007; Fowler and Christakis, 2010). Unlike direct reciprocity, it requires no repeated exchange with the same person: it operates as a pay-it-forward principle, sustaining collective contributions in groups — such as online knowledge-sharing platforms — where interactions are infrequent and non-recurring.

Socio-psychological research identifies two complementary mechanisms underlying generalized reciprocity: moral reasoning and emotional response. On the moral side, individuals who reason by universalization — asking what would happen if all members requested more help than they gave — conclude that widespread free-riding would undermine the community as a source of help, generating a shared obligation to reciprocate toward the group (Levine et al., 2020; Curry et al., 2019). On the emotional side, gratitude aroused by the perceived generosity of others motivates prosocial action beyond direct reciprocation (McCullough et al., 2001; Tsang, 2006; Bartlett and DeSteno, 2006) and is especially relevant where free riding is easy, as on knowledge-sharing platforms (Molm et al., 2007; Elster, 2009). These moral and emotional pathways are the *proposed* generative mechanisms behind generalized reciprocity; our data capture the helping behavior they predict, not the mechanisms themselves.

Evidence that generalized reciprocity can operate — and can initiate cascading acts of generosity (Fowler and Christakis, 2010; Boyd and Richerson, 1989) — is robust: comparative research indicates that the underlying norm is globally shared (Curry et al., 2019), and experimental demonstrations span laboratory (Greiner and Levati, 2005; Molm et al., 2007; Stanca, 2009) and field settings (Mujcic and Leibbrandt, 2018; Fowler and Christakis, 2010).

The open question is how strongly it operates among real users on large platforms — a question observational data have answered inconsistently, for reasons we argue are largely methodological.

2.2. Empirical Challenges in Measuring Generalized Reciprocity

One challenge is the reliance on self-reported motives in surveys and interviews. Contributors frequently describe reciprocity as a central reason for their engagement (Wasko and Faraj, 2000; Lakhani, 2003), yet people have difficulty articulating the moral motives behind their actions (Vaisey, 2009; McClelland et al., 1989), and introspection struggles to separate reciprocity from altruism, conformity to perceived norms, status pursuit, or the popular rhetoric of “giving back” as post-hoc rationalization (Batson, 1991; Bénabou and Tirole, 2006). Self-reported data may therefore overstate the prevalence of reciprocity as a discrete motive.

The more fundamental challenge is endogeneity: the people who ask for help are also the people most likely to give it. Users who ask questions on platforms like Stack Overflow answer them at disproportionately high rates — not because receiving help triggered a reciprocal response, but because both behaviors reflect the same underlying characteristics, such as greater platform engagement. Studies that correlate help received with help given without accounting for this baseline activity effectively confound reciprocity with user type (Baker and Bulkley, 2014; Chen et al., 2018; Joyce and Kraut, 2006). This confound is the central identification challenge that motivates the matched difference-in-differences design we develop in Section 4.

Prior observational studies of online reciprocity report mixed magnitudes. Baker and Bulkley (2014) and Chen et al. (2019) find positive associations between help received and subsequent helping; Wasko and Faraj (2005) and Joyce and Kraut (2006) find weak or null relationships. Our matched DiD estimates sit below the larger positive reports, consistent with the view that designs that do not net out baseline activity overstate the effect. The tenure gradient — clear elevation among newcomers, null among veterans — offers one reconciliation of the mixed prior findings.

2.3. Reciprocity and the User Trajectory

We propose that the mixed findings in the literature can be partly reconciled by attending to a dimension that prior research has largely overlooked: the user trajectory from newcomer to experienced contributor. We argue that

generalized reciprocity is strongest for new users and attenuates as users gain experience on the platform.

The logic of this prediction draws on theories of socialization and norm internalization. When a user first encounters an online community, they are unfamiliar with its specific norms, social structures, and reward systems. At this stage, community-specific incentives, such as reputation points, badges, or leaderboard positions, carry little motivational weight because the newcomer has not yet learned to value them. Instead, initial contributions are guided by more general moral principles that operate independently of community-specific knowledge. Generalized reciprocity is precisely such a principle: it requires only that the individual has received help and feels a moral obligation or emotional impulse to help in return, without any need to understand or identify with the particular community (Levine et al., 2020; Curry et al., 2019).

As users interact with the community, they gradually learn its norms and develop a sense of membership. On platforms like Stack Overflow, these norms are codified in explicit reputation systems that reward helpful behavior with visible status markers. Once users internalize these community-specific norms, the broad moral obligation of reciprocity may be displaced by more localized incentive structures. The logic of universalization (“I should help because I was helped”) gives way to the logic of community membership (“I should help because that is what valued members do”). This norm displacement echoes accounts of how external incentives can crowd out moral motives (Bénabou and Tirole, 2006).

Two further mechanisms accelerate this decline. As users accumulate experiences of receiving help, each additional instance becomes less surprising, and the gratitude that fuels reciprocity habituates (McCullough et al., 2008). And as users learn that the platform rewards helping with reputation and status, they may increasingly attribute others’ helpfulness to status pursuit rather than genuine generosity, further dampening that gratitude.

This account predicts that generalized reciprocity is strongest among newcomers and declines with experience. It also helps explain why prior studies have reached different conclusions about the importance of reciprocity: those that focus on or oversample experienced users may underestimate reciprocity, while those that capture newcomer behavior may find stronger effects.

HYPOTHESIS 1 (H1). *Receiving help increases a user’s propensity to help others (generalized reciprocity effect).*

HYPOTHESIS 2 (H2). *The generalized reciprocity effect is strongest among newcomers and decreases as users gain experience on the platform.*

The stakes of this prediction extend beyond reconciling mixed findings. If the reciprocity association is concentrated among newcomers, generalized reciprocity would function less as a general engine of cooperation among established members than as a *contributor-recruitment* channel operating at the point of entry, before community-specific incentives take hold — a design implication we develop in Section 6.4.

2.4. Response Time and the Immediacy of Reciprocity

The emotional mechanisms underlying reciprocity suggest that the speed of help matters for its strength. Gratitude, the primary proposed emotional driver of generalized reciprocity (McCullough et al., 2001), is triggered by the perception of a benefactor’s generosity, and its intensity is likely to depend on the characteristics of the helping experience — including its timeliness.

A swift response can amplify the experience of receiving help: a fast answer signals that the community is attentive and generous, and it arrives while the asker’s need is still salient, when gratitude should be most intense. Conversely, when an answer arrives hours or days later, the original need may have faded or been met elsewhere, the associative link between asking and receiving weakens, and the delay itself may signal an unresponsive community — attenuating the emotional and moral responses that underpin reciprocation.

These considerations lead to a straightforward prediction:

HYPOTHESIS 3 (H3). *Timelier responses to a user’s question strengthen the generalized reciprocity effect. Laboratory accounts of gratitude imply that this advantage grows monotonically as help arrives sooner; because field settings add session dynamics those accounts abstract away, we treat the prediction as directional and test whether it holds monotonically.*

This hypothesis follows from the dominant theoretical accounts of generalized reciprocity, which emphasize emotional immediacy and the salience of the helping experience. However, as we will show, field settings introduce session dynamics — whether the asker is still online when the answer arrives — that qualify the predictions of laboratory-based theory. We return to this issue when presenting our results.

3. Empirical Setting: Stack Overflow

Stack Overflow is well suited to examining generalized reciprocity in on-line knowledge sharing. The platform publishes quarterly data dumps that record what every user did and when, reaching back to its inception. From these records we can reconstruct how people helped one another over time, and in what order they received and gave help — the sequence on which any study of generalized reciprocity depends. Stack Overflow’s active question-and-answer volume has declined sharply since the release of generative AI tools in late 2022; our sample spans platform inception through 1 April 2025 and therefore covers a period that mostly predates, but extends into, the steepest LLM-driven decline. We treat the findings as a historical snapshot of one large, technical, relatively anonymous, status-driven platform; external validity to other communities is discussed in Section 6.

At its core, Stack Overflow is a question-and-answer platform for software developers and technologists. As of mid-2025, the site had accumulated over 24 million questions and 36 million answers from more than 4.5 million users. Its user base spans a wide range of experience levels, from learners to professional software developers (Stack Overflow, 2024). Questions on the platform must be tightly focused on specific programming problems; broad or opinion-based inquiries are typically closed by community moderation (Stack Exchange Inc., 2024). The most frequently discussed topics, as reflected in question tags, include JavaScript, Python, Java, C#, PHP, Android, and HTML, spanning areas from web development and mobile applications to data science and machine learning. Stack Overflow is built around several core features: tags categorize questions by topic, facilitating expertise matching; a voting system allows users to upvote or downvote questions and answers to assess quality; and reputation points are earned for valued contributions such as providing high-quality answers (Stack Exchange Inc., 2024).

For our study, the key feature is the question-and-answer interaction between users. When a user posts a question, other users may provide answers. This creates a timestamped sequence (question posted, answer received) that we exploit to measure the temporal dynamics of generalized reciprocity. On Stack Overflow, question-askers and answerers are typically different individuals, which makes the platform well suited to studying *generalized* reciprocity: users who receive help from one community member may subsequently help entirely different members.

4. Data and Methods

To test our predictions, we develop a matched staggered difference-in-differences survival analysis that exploits the order in which users ask for and give help on Stack Overflow. Our design compares changes in helping behavior (measured as the hazard of posting an answer) before and after a user receives an answer, between users who receive an answer and matched users who do not. This approach integrates two methodological traditions: the staggered difference-in-differences framework of Callaway and Sant’Anna (2021), which accommodates variation in treatment timing and treatment effect heterogeneity, and the counting-process formulation of the Cox proportional hazards model (Andersen and Gill, 1982), which allows treatment status and other covariates to vary continuously over the observation window.

4.1. Study Design

For each question posted by a user, we construct a timeline centered on two key events: the posting of the question (T_{question}) and, for questions that receive an answer, the time at which the first answer with a non-negative vote score is posted (T_{answer}). The observation window opens two days before the question ($T_{\text{question}} - 2D$) and closes two days *after the answer arrives* ($T_{\text{answer}} + 2D$). The window is therefore anchored on the answer rather than on the question: the pre-question and post-answer segments each have a fixed two-day duration, while the intervening waiting period runs for the question’s response time. The total timeline spans $4D + RT$ (response time), so slower-answered questions have proportionally longer timelines; this ensures every treated question — however late its answer — receives a full two-day post-answer phase in which the reciprocity effect can be observed. The timeline is divided into distinct phases.

We define three phases within each observation window (see Figure 1). The first phase, the *pre-question phase*, is from the start of the observation window to the posting of the question ($T_{\text{question}} - 2D$ to T_{question}). This phase captures baseline helping activity before the focal question is asked. The second phase, the *waiting period*, is from the posting of the question to the receipt of the first answer (T_{question} to T_{answer}). During this interval, the user has asked for help but has not yet received it. The final *post-answer phase* is from the receipt of the answer to the end of the observation window (T_{answer} to $T_{\text{answer}} + 2D$), a fixed two-day span regardless of how long the answer took

to arrive. This is the phase in which the reciprocity effect, if it exists, should manifest.

A key feature of our design is that the post-answer (treated) phase begins at different elapsed times across observations, depending on how quickly the posted question receives a response.

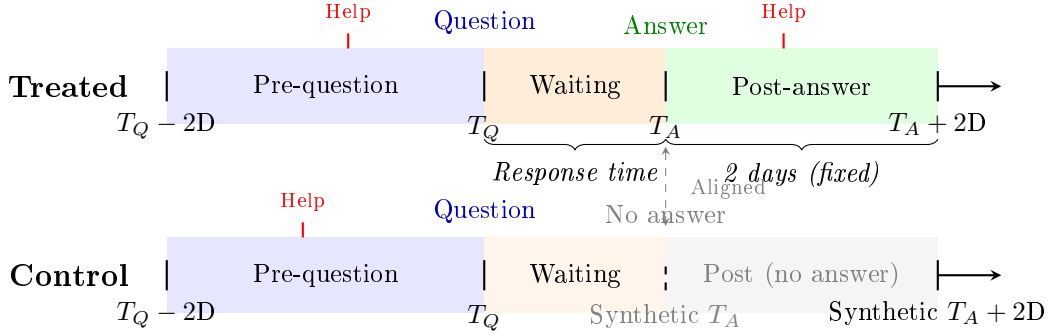
This staggered timing of the post-answer phase corresponds to the setting studied by Callaway and Sant’Anna (2021), who show that standard two-way fixed-effects estimators can be biased when treatment timing varies, and treatment effects are heterogeneous. Our approach avoids this pitfall by explicitly modeling the time-varying change in treatment status within each observation window and by comparing treated observations only to not-yet-treated or never-treated controls.

For control users (those whose questions did not receive an answer), there is no natural transition between the waiting period and the post-answer phase. To maintain temporal alignment with the matched treated observation, we assign the control observation a synthetic phase transition at the same elapsed time as the treated user’s response time. That is, if the treated user’s answer was posted 4 hours after their question, the matched control observation’s timeline is split at $T_{\text{question}} + 4\text{h}$ and its window closes at $T_{\text{question}} + 4\text{h} + 2\text{D}$, creating an aligned “post” phase of the same two-day duration. Because the control inherits the treated response time, matched treated and control observations share identical window boundaries and identical two-day post-answer durations, so any differences between them in the post-answer phase are not driven by differences in the length of the observation period.

The survival clock in our Cox models is centered on posting of the respective question: time zero corresponds to the start of the observation window ($T_{\text{question}} - 2\text{D}$), so that the question itself is posted at day 2 of the timeline. Because the window closes two days after the answer, its end falls at day $4 + \text{RT}$ rather than at a fixed day 4; a question answered within the first day therefore closes near day 4, while a question answered after three days closes near day 7. All help events (answers posted by the focal user) are recorded with their exact timestamps and converted to elapsed time on this clock.

Within each timeline, we track all instances in which the user provides an answer to another user’s question. These helping events constitute the outcome in our survival analysis; broader outcome definitions that also count comments and edits are examined in the decomposition (Table 6). Conceptu-

Figure 1: Study Design: Matched Difference-in-Differences Survival Analysis



Note. The observation window opens two days before the question (T_Q) and closes two days after the answer ($T_A + 2D$); it is anchored on the answer, so the pre-question and post-answer segments each last a fixed two days while the intervening waiting period spans the response time. The total window length is thus $4D + RT$ and is not drawn to scale here. The treated timeline (top) shows three phases: the pre-question phase (baseline), the waiting period (question posted, no answer yet), and the post-answer phase (after the first qualifying answer T_A arrives). The control timeline (bottom) is from a propensity-score-matched user whose question did not receive an answer. To ensure temporal alignment, the control's post-question period is split at a synthetic transition point matching the treated user's response time. Red markers indicate help events (answers posted by the focal user to other users' questions). The treatment effect is the differential change in helping hazard between treated and control observations in the post-answer phase, relative to the pre-question baseline. The staggered timing of T_A across observations motivates our use of matched DiD.

ally, our design asks: Does the hazard of helping increase after a user receives an answer, relative to the change in hazard for a matched user who does not receive an answer? We present descriptive evidence on the raw help rates over this timeline in Section 5 (Figure 2), before turning to the formal Cox model estimates.

4.2. Data

We used the full sample of non-deleted questions from Stack Overflow’s inception on 31 July 2008 through 1 April 2025. We addressed two edge cases that would render our fixed-length observation window invalid. First, for early platform questions, the pre-period could predate the platform’s launch, meaning no answers were possible at that time. We removed 1,080 such cases. Second, due to the data dump’s 1 April 2025 cutoff, the post-period could be prematurely truncated. We excluded 8,029 cases affected by this. We then excluded 2,671,121 questions that received an answer from the question poster. These filtering steps resulted in a pre-matching dataset of 21,033,429 questions from 4,764,048 unique users. After propensity score matching (described below), the matched file comprises 24,349,760 question-level rows (12,174,880 matched pairs; some control questions appear more than once) from 3,732,706 unique users, with a 50% treatment rate by construction. All matched rows enter the Cox analysis; the helping outcome is the posting of an answer to another user’s question (answers to one’s own question do not count), yielding 3,435,973 helping events in the pooled analysis sample (Table 5).

4.3. Variables

Table 1 provides an overview of the variables used in our analysis. The phase terminology is consistent with Figure 1: the pre-question phase captures baseline behavior, the waiting period spans the interval from question to answer, and the post-answer phase captures the period after the answer arrives. We use “post-question” to refer to the combined waiting period and post-answer phase whenever distinguishing between these two sub-periods is not necessary.

Outcome. The event of interest is the posting of an answer to another user’s question — the platform’s central helping act. Within each window we track when (and whether) the focal user answers any other user’s question, and the Cox model estimates the instantaneous hazard of this helping event as a

Table 1: Overview of Variables

Variable	Description
<i>Outcome (Survival Event)</i>	
helpEvent	A binary event indicator equal to 1 at times when the user posts an answer to another user’s question within the observation window (answers to one’s own question do not count). Broader definitions adding comments and edits are reported in the outcome decomposition.
<i>Time-Varying Covariates</i>	
PostQuestion	Binary; 1 for intervals occurring after the user posts their question ($t \geq T_{\text{question}}$), 0 otherwise. Captures the combined post-question period (waiting period + post-answer phase).
PostAnswer	Binary; 1 for intervals occurring after the user receives their first answer ($t \geq T_{\text{answer}}$), 0 otherwise. For control users (no answer received), this variable switches on at the synthetic transition point at the matched treated user’s response time. Isolates the post-answer phase.
treatedPostQuestion	Interaction of treatment $\times$ phasePostQuestion; captures differential behavior of treated users in the entire period after the question.
isTreatedActive	Interaction of treatment $\times$ phasePostAnswer; the primary treatment effect capturing the additional lift in helping hazard after receiving an answer, beyond any change attributable to the act of asking a question.
ResponseTime	Time between question and the first answer (excluding down-voted answers) of the treated question. We include interactions with all four variables above.

function of time-varying covariates that capture the phase of the observation timeline and the treatment status. Answering one’s own question, and self-directed acts such as accepting an answer, do not count as helping events. To address the concern that answers alone may understate lower-cost reciprocity, we re-estimate the pooled model under broader outcome definitions — comments only, edits only, and composites that add these channels to answers — and report the full decomposition in Table 6.

Why answers anchor the construct.. We treat the posting of an answer as the theoretically central helping act. Generalized reciprocity, as modeled in the experimental literature, is a *costly* prosocial contribution: individuals pass a benefit to an unrelated third party at real expense to themselves (Greiner and Levati, 2005; Stanca, 2009; Mujcic and Leibbrandt, 2018), and it is the willingness to bear that cost — not mere approval or endorsement — that distinguishes reciprocity from low-cost signaling (Molm et al., 2007). Answering a stranger’s question is the platform analogue: it demands substantive effort and yields a public good for the community. Lower-cost gestures such as comments, upvotes, and edits index engagement, curation, or approval more than the effortful contribution the construct denotes. We therefore anchor the outcome on answers and report comment- and edit-inclusive alternatives in the decomposition (Table 6); upvotes are the one lower-cost channel our processed data do not record. As the decomposition shows, this choice does not manufacture the effect: comments alone carry only a slight elevation, well below the answers increment, so broadening the outcome dilutes rather than inflates the estimate (see Section 6.5).

Treatment.. The treatment is the receipt of an answer. We measure it with *treatment*, a binary variable set to one when the user’s question received, within seven days of posting, at least one answer from *another* user carrying a non-negative score; the seven-day cap keeps the fixed two-day post-answer window inside the observable record. Control questions are the complement — those that received no such qualifying answer — which pools three cases that share the absence of timely, positively scored help from another user: questions never answered, questions whose only answers within seven days were downvoted, and questions whose first qualifying answer arrived only after seven days. We key treatment on receiving an answer rather than on Stack Overflow’s “accepted answer” mechanism because accepting an answer requires the user to return to the site, and users who return are plausibly also the users who go on to help.

Response Time. For each treated observation, the response-time variable used in the interaction terms is $\log(1 + \text{RT})$, where $\text{RT} = t_{\text{answer}} - t_{\text{question}}$ is the actual response time in hours: the elapsed time from the posting of the question to the arrival of the first answer with a non-negative vote score. The origin is the question itself, so RT is the true question-to-answer latency and is unaffected by the length of the pre-question window. The discrete response-time bins reported below (Table 9) are likewise cut on this same actual response time, so a bin labelled “1–3 days” contains questions answered 24–72 hours after posting. For numerical stability, both response-time interaction terms ($\text{isTreatedActive} \times \log(1 + \text{RT})$ and $\text{treatedPostQuestion} \times \log(1 + \text{RT})$) are standardized (z-scored) and then clipped to their 5th–95th percentile range before model fitting. Consequently, the reported interaction coefficients γ and δ are expressed per one standard deviation of the respective interaction term rather than per unit of $\log(1 + \text{RT})$.

User Experience. To test Hypothesis 2, we stratify our analysis by user tenure, defined as the number of days between a user’s first activity on the platform and the start of the observation window. We define seven tenure buckets using right-closed intervals: $[0, 7]$, $(7, 30]$, $(30, 180]$, $(180, 365]$, $(365, 1095]$, $(1095, 2190]$, and $(2190, \infty)$ days, corresponding to the labels less than one week, one week to one month, one to six months, six to twelve months, one to three years, three to six years, and more than six years, respectively. Running separate models by bucket enables us to examine how the reciprocity effect varies across the user trajectory without imposing a parametric functional form.

4.4. Propensity Score Matching

Because the receipt of an answer (the treatment) is correlated with user characteristics (more experienced users, for instance, might tend to ask higher-quality questions that are more likely to receive answers), we employ propensity score matching (Rosenbaum and Rubin, 1983) to improve the comparability of the treatment and control groups. Importantly, matching is performed *across different users*: each treated question (from a user who received an answer) is matched to a control question (from a different user whose question did not receive an answer) that has a similar propensity to receive an answer based on observable pre-treatment characteristics. This cross-user matching ensures that treated and control observations are independent conditional on the matched covariates, while the DiD structure

within each pair absorbs residual differences.

We estimate the propensity score using pre-treatment covariates that jointly capture user experience, recent activity, and question characteristics, as listed in Table 2. The covariate *tagAcceptShareAvg* addresses the concern that question complexity may confound the treatment: questions in tags with low accepted-answer rates are inherently more difficult, and matching on this rate ensures that treated and control questions come from similarly tractable topic areas. Matching on the tag-level accepted-answer rate also matters for the response-time analysis, because it mitigates the concern that slower response times simply proxy for differently sized sub-communities with lower throughput. We use nearest-neighbor matching *with replacement* and a caliper of 0.05 on the propensity score, requiring in addition an exact match on the calendar year, the question’s primary tag, and the number of tags. Because the pool of unanswered questions is smaller than the pool of answered questions, some control questions are matched to more than one treated question; the matched file therefore contains more rows than unique question pairs (12.2 million pairs, 24.3 million rows). Each row enters the Cox analysis as an independent observation window; standard errors are discussed in the robustness appendix.

Table 3 displays the covariate balance before and after matching. Before matching, several covariates showed a meaningful imbalance between the treated and control groups. For instance, treated users had asked more questions in the prior 30 days ($SMD = 0.197$) and 7 days ($SMD = 0.174$), reflecting that users who ask questions that attract answers tend to be more recently active; their questions were also longer and posted by higher-reputation askers (body length $SMD = -0.183$; owner reputation $SMD = 0.115$). The largest imbalance was in the tag-level accepted-answer rate ($SMD = 0.695$), confirming that question topic characteristics strongly predict answer receipt and underscoring the importance of matching on this variable. The propensity model addresses observable selection directly by including posting hour and day, question and title length, and owner reputation (Table 2). After matching, all standardized mean differences fell well below the conventional threshold of 0.1, with the largest residual imbalance below 0.05 (see Table 3). This confirms adequate balance across all matching covariates; the full set, including posting-time features, is illustrated in the Love plot in Appendix Appendix B.

Table 2: Definition of Matching Covariates

Variable name	Definition
<i>Exact matching</i>	
calendarYear	Calendar year exact matching indicating the year of the observation, controlling for secular trends in platform activity and answer rates.
topLevelTag	First tag of the question, indicating the main topic and the community that sees the question.
numTags	Number of tags on the question. Treated and control questions are matched on the exact tag count, so they pose comparably scoped problems.
<i>Propensity score matching</i>	
userTenure	Number of days since the user’s first recorded activity on the platform, capturing cumulative experience.
numQuestionsAskedAT	Total number of questions asked by the user prior to the observation window.
numHelpProvidedAT	Total number of answers provided by the user prior to the observation window.
numQuestionsAsked30D	Number of questions asked in the 30 days preceding the observation window.
numHelpProvided30D	Number of answers provided in the 30 days preceding the observation window.
numQuestionsAsked7D	Number of questions asked in the 7 days preceding the observation window.
numHelpProvided7D	Number of answers provided in the 7 days preceding the observation window.
tagAcceptShareAvg	Average share of questions that receive an <i>accepted</i> answer within the question’s tags, capturing topic-specific difficulty.
<i>Observable question and author characteristics (propensity score matching)</i>	
postHour	Hour of day at which the question was posted, capturing diurnal variation in who is online to answer.
postDayOfWeek	Day of week on which the question was posted.
bodyLenChars	Length of the question body in characters, a proxy for question detail and effort.
titleLenChars	Length of the question title in characters.
ownerReputation	Asker’s reputation score at posting, capturing standing and experience beyond raw tenure.

Table 3: Covariate Balance Before and After Matching

Covariate	Unmatched			Matched		
	Tr Mean	Ct Mean	SMD	Tr Mean	Ct Mean	SMD
userTenure	570.95	720.62	−0.167	568.37	544.76	0.026
numQuestionsAskedAT	31.11	26.00	0.059	30.34	28.04	0.027
numHelpProvidedAT	15.27	14.15	0.009	14.36	12.15	0.018
numQuestionsAsked30D	2.00	1.19	0.197	1.96	1.84	0.030
numHelpProvided30D	0.81	0.46	0.068	0.76	0.62	0.028
numQuestionsAsked7D	0.57	0.34	0.174	0.56	0.53	0.023
numHelpProvided7D	0.22	0.12	0.063	0.20	0.16	0.026
tagAcceptShareAvg	0.51	0.44	0.695	0.51	0.51	−0.003
numTags	2.83	3.04	−0.173	2.83	2.83	0.000
bodyLenChars	1289.10	1658.54	−0.183	1286.37	1338.86	−0.026
titleLenChars	53.80	57.50	−0.172	53.79	53.97	−0.009
ownerReputation	4300.57	2541.81	0.115	4130.42	3499.00	0.041

Note. SMD = standardized mean difference. The conventional threshold for acceptable balance is $|\text{SMD}| < 0.1$. All post-matching SMDs fall well below this threshold.

4.5. Analytical Strategy: Cox Proportional Hazards with Time-Varying Covariates

We model the hazard of helping using a Cox proportional hazards model with time-varying covariates. The counting-process formulation of Andersen and Gill (1982) extends the original Cox model (Cox, 1972) to settings with time-varying covariates and recurrent events, providing the large-sample theoretical foundation (including consistency and asymptotic normality of the partial likelihood estimator) that justifies our approach.

For each matched pair of observations (one treated, one control), we partition the 4-day observation window into intervals defined by the key event times: the start of the window, the posting of the question, and the arrival of the answer (the actual answer arrival for treated users or the synthetic transition for controls).

The hazard function is specified as:

$$h(t \mid \mathbf{X}(t)) = h_0(t) \exp \left(\beta_0 \cdot \text{treatment} + \beta_1 \cdot \text{phasePostQuestion}(t) + \beta_2 \cdot \text{treatedPostQuestion}(t) \right. \\ \left. + \beta_3 \cdot \text{phasePostAnswer}(t) + \beta_4 \cdot \text{isTreatedActive}(t) \right) \quad (1)$$

where $h_0(t)$ is the baseline hazard. *treatment* is a time-constant indicator for treatment group membership (1 if the user’s question received an answer, 0 otherwise); it absorbs stable between-group differences in baseline helping rates. The remaining covariates are time-varying indicators that switch on at the appropriate event times. Because *treatedPostQuestion* switches on when the question is posted and remains on through the post-answer phase, the treated–control gap in the helping hazard equals β_2 during the waiting period and $\beta_2 + \beta_4$ after the answer arrives. The coefficient β_4 is therefore the difference-in-differences estimate of the change at answer arrival: the shift in the treated–control gap at the moment the answer lands, net of whatever divergence preceded it. We report β_4 as the treatment effect. The waiting-period coefficient β_2 is reported separately as a parallel-trends diagnostic: because no answer has yet arrived during the waiting period, it cannot reflect a response to answer receipt — it measures anticipatory or selection-driven divergence, carries no sign guarantee, and is accordingly not folded into the effect estimate.

To test the moderation by response time (Hypothesis 3), we extend the model with two additional terms:

$$h(t \mid \mathbf{X}(t)) = h_0(t) \exp \left(\beta' \mathbf{X}(t) + \gamma \cdot \text{isTreatedActive}(t) \times \widetilde{\log}(\text{RT}) + \delta \cdot \text{treatedPostQuestion}(t) \times \widetilde{\log}(\text{RT}) \right) \quad (2)$$

where $\widetilde{\log}(\text{RT})$ denotes the standardized log response-time variable (see below), γ captures the moderation of the answer-arrival increment by answer speed, and δ the analogous moderation during the waiting period. The logic mirrors the base model exactly. Just as β_4 alone — not the sum $\beta_2 + \beta_4$ — is the treatment effect, γ alone is the response-time moderation of that effect: it scales the arrival increment β_4 , the difference-in-differences contrast on which the reciprocity effect rests. We therefore test H3 on γ . A negative γ indicates that faster answers strengthen the arrival increment, the direction H3 predicts; a positive γ indicates the opposite. The waiting-period moderation δ is the counterpart of β_2 : because it scales a pre-answer gap that cannot reflect a response to an answer not yet received, it carries no sign guarantee, and we report it separately as a diagnostic rather than folding it into the H3 test.

Stratification by Tenure. To test Hypothesis 2, we fit separate models for each of the seven tenure buckets. This allows the baseline hazard and all coefficients to vary freely across levels of user experience, providing a non-parametric view of how reciprocity changes over the user trajectory.

Identification. Our identification strategy rests on the matched DiD logic embedded in the model. Propensity score matching (Rosenbaum and Rubin, 1983) ensures that treated and control observations come from users with comparable baseline characteristics, including tenure, activity levels, and question topic difficulty. This matching absorbs between-user differences that might otherwise confound the treatment effect.

We do not include user fixed effects in the survival models. Because matching is performed across different users (not within-user), and because the Cox model already leverages within-observation temporal variation through the time-varying covariate structure (Andersen and Gill, 1982), user fixed effects are not required for identification. The DiD structure compares within-observation changes before and after the answer across treatment groups, isolating the treatment effect from stable individual differences.

The waiting-period covariate (*treatedPostQuestion*) captures any lift in helping that treated users show after asking but before receiving an answer. When this term is near zero, treated and control users move in parallel up to the answer, and the arrival increment β_4 can be read as the response to answer receipt rather than to the act of asking itself. When it is not, the two groups diverge before treatment arrives; β_4 still nets out the average waiting-period gap, but we flag such strata and interpret their estimates with added caution rather than treating the divergence as part of the effect. Even a near-zero waiting-period term, however, rules out only motivational shifts that materialize *before* the answer. Question-induced channels that operate mainly after the answer arrives are not absorbed by it — most notably the elevated profile visibility a question generates (see Section 6.5). We therefore report the waiting-period term explicitly, so that departures from parallel trends are measured rather than assumed away, and we return to its magnitude in the Results.

5. Results

5.1. Descriptive Statistics

We analyze 24,349,760 question-level observations from 3,732,706 users on Stack Overflow, half of whose questions received at least one answer (a

50% rate fixed by construction of the matched sample, not a platform base rate). Most users help no one in a given window: they post 0.18 answers to other people’s questions on average (SD = 1.24; median 0), so helping is rare for any individual user over any four-day stretch. Half of the users who got an answer waited less than 0.34 hours for it, though a minority waited far longer (mean 5.08 hours, SD = 17.06). Our users span every level of platform experience. The largest group had been on the platform one to three years ($n = 6,424,968$), followed by those at one to six months ($n = 4,045,952$) and those in their first week ($n = 4,029,008$).

Table 4: Descriptive Statistics

Variable	Mean	SD	Median	N
Questions (observations)				24,349,760
Unique users				3,732,706
% questions with answer	50.00%			
Help events per window (answers to others)	0.18	1.24	0.00	
User tenure (days)	656	839	328	
Response time (hours, treated)	5.08	17.06	0.34	
<i>Response time (hours, treated) by Tenure Bucket</i>				
< 1 Week	4.51	15.73	0.32	2,014,406
1 Week - 1 Month	3.59	13.83	0.26	775,751
1 - 6 Months	3.83	14.45	0.26	2,022,884
6 - 12 Months	4.24	15.42	0.28	1,491,553
1 - 3 Years	5.11	17.17	0.34	3,212,434
3 - 6 Years	6.69	19.87	0.50	1,807,077
> 6 Years	8.66	22.72	0.75	850,441
<i>N (unique questions) by Tenure Bucket</i>				
< 1 Week				4,029,008
1 Week - 1 Month				1,551,586
1 - 6 Months				4,045,952
6 - 12 Months				2,983,180
1 - 3 Years				6,424,968
3 - 6 Years				3,614,184
> 6 Years				1,700,882

5.2. Descriptive Evidence: Help Rates and the Parallel Trends Assumption

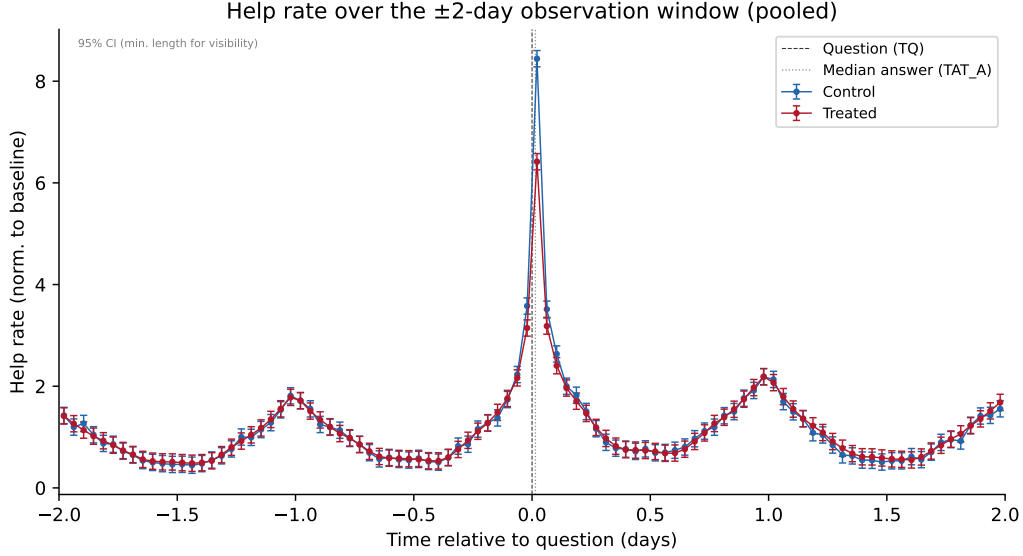


Figure 2: Help rate over a fixed ± 2 -day window centered on question posting, pooled across all tenure buckets. For legibility this descriptive figure uses a symmetric question-centered axis ($T_Q - 2D$ to $T_Q + 2D$); it is not the answer-anchored analysis window of Figure 1, which closes two days after the answer ($T_A + 2D$) and is the window used in the Cox models. The normalized help rate (relative to each group’s pre-question baseline) is plotted for treated users (red, who received an answer) and matched control users (blue, who did not). The dashed vertical line marks the question time (T_Q) and the dotted line the median answer arrival; error bars indicate 95% confidence intervals.

Figure 2 plots the normalized help rate for treated and control users over the two days before and after question posting. Before the question ($t < T_Q$), treated and control help rates track each other closely, consistent with the parallel-trends assumption and with the effectiveness of propensity score matching. At the moment of posting, both groups’ help rates spike sharply — users who have just asked a question are, mechanically, online and active — and the spike is more pronounced for controls, who continue waiting for an answer, than for treated users, whose questions are typically answered within the hour. Beyond this mechanical spike, the treated–control differences implied by our estimates (a few percent of the helping hazard) are too small to be visible in raw normalized rates, which is why the model-based difference-in-differences estimates below are needed to resolve them.

We return to the control group’s larger at-question spike when discussing the response-time pattern.

5.3. *Generalized Reciprocity Effect (H1)*

To test Hypothesis 1, we estimate the pooled Cox model across all experience levels (Table 5). The treatment effect is the difference-in-differences increment at answer arrival (β_4); the waiting-period term (β_2) is reported alongside it as a parallel-trends check. Users who received an answer went on to help others slightly more often than their matched counterparts did, and the gap is small: the arrival increment is $\text{HR} = 1.05$ (95% CI [1.04, 1.05], $p < .001$), roughly a 5% higher rate of helping once the answer lands. The waiting-period term is small and negative ($\beta_2 = -0.007$, $p = .01$): distinguishable from zero at this sample size, but an order of magnitude smaller than the arrival increment and opposite in sign. Treated users did not diverge upward from controls before the answer arrived; if anything, they helped slightly less. The arrival increment is therefore not inflated by a pre-answer trend. Matched-pair bootstrap uncertainty for the pooled model, which preserves the matched structure, is reported in Appendix Table D.11 and confirms the model-based interval. Hypothesis 1 thus receives support in a qualified form: the association is far smaller than prior observational reports, and it remains subject to the answer-quality caveat developed below (Section 6.5).

A waiting-period term this close to zero indicates that asking a question did not, by itself, change how much treated users helped while they waited. It does not, however, rule out influences that the question sets in motion and that persist after the answer arrives, such as the extra profile visibility a question attracts (see Section 6.5). This modest pooled figure also hides a great deal of variation: newcomers respond most strongly, and long-tenured users not at all. Before turning to that heterogeneity, we decompose the outcome by help type.

Table 5: Pooled Cox Regressions (All Experience Levels)

	Main	Main + Speed
<i>Treatment effect (difference-in-differences): increment at answer arrival (β_4)</i>		
Answer arrival ($\times$ Post-Answer Received)	0.0479*** (0.0026)	0.0373*** (0.0022)
Hazard Ratio [95% CI]	[1.04, 1.05]	[1.03, 1.04]
Hazard Ratio [bootstrap 95% CI]	[1.03, 1.07]	—
<i>Parallel-trends diagnostic: waiting period (β_2)</i>		
Waiting period (β_2): Received Answer $\times$ Post-Question	-0.0065* (0.0026)	0.0142*** (0.0021)
<i>Response Time Interaction</i>		
Answer arrival $\times$ log(RT) (γ)	—	-0.0636*** (0.0015)
Waiting period $\times$ log(RT) (δ)	—	0.0630*** (0.0013)
N	24,349,760	24,349,760
Events	3,435,973	3,435,973

Notes: Matched-pair bootstrap 95% CI for the pooled arrival increment is from Table D.11.

Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; $^\dagger p < 0.1$

In response to the concern that an answers-only outcome may understate reciprocity by excluding lower-cost helping acts, Table 6 re-estimates the pooled model under broader outcome definitions. The decomposition confirms that the reciprocity signal is concentrated in *answers*. Comments alone show only a slight elevation (arrival HR = 1.01, 95% CI [1.01, 1.02]), well below the answers-only increment, and adding comments to answers dilutes rather than amplifies the estimate (answers-plus-comments arrival HR = 1.02, [1.02, 1.03]). That is the opposite of what a mechanical undercount of low-cost reciprocity among novices would produce. Edits of other users' posts carry a larger association (edits-only arrival HR = 1.16), but we report them separately because they index curation more than costly contribution; the all-channel composite sits between the two (answers-plus-comments-plus-edits arrival HR = 1.05). The waiting-period diagnostic is near zero in every column except edits ($\beta_2 = +0.06$), and the effect enters at answer arrival, mirroring the pooled pattern. The answers-only estimate we headline is thus not an artifact of ignoring low-cost channels: the channel

closest to conversational helping (comments) contributes little, and the one channel that would raise the estimate (edits) indexes a different construct.

Table 6: Reciprocity Effect Decomposed by Help Type

Outcome	Arrival HR [95% CI] (β_4)	Waiting β_2	Events
Answers only	1.05 [1.04, 1.05]	-0.01	3,435,973
Comments only	1.01 [1.01, 1.02]	+0.01	2,170,711
Edits only	1.16 [1.16, 1.17]	+0.06	992,369
Answers + comments	1.02 [1.02, 1.03]	-0.00	5,025,482
Answers + comments + edits	1.05 [1.04, 1.05]	-0.00	5,679,602

Notes: Accepting an answer is excluded from the decomposition: it is a self-directed act available only to treated users (a control never receives an answer to accept), so the matched difference-in-differences contrast is degenerate for it.

Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

5.4. Experience Moderation of Reciprocity (H2)

To test Hypothesis 2, we estimate the model separately within each of the seven tenure buckets (Table 7; Figure 3). The response is strongest at entry and fades to nothing among the most experienced, though the decline across the middle of the tenure range is not strictly monotone. The newest users respond most: in their first week, users who received an answer went on to help roughly 7.1% more often than their matched counterparts (arrival HR = 1.07, 95% CI [1.06, 1.09], $p < .001$). The response then dips in the one-week-to-one-month bucket (HR = 1.03 [1.01, 1.05]), recovers to a plateau of about 1.05 that holds from one month through three years, and weakens again at three to six years (HR = 1.02 [1.01, 1.04], $p < .001$). Only users with more than six years on the platform show no response at all (HR = 1.00 [0.98, 1.02], n.s.). Read at this level — elevated at entry, roughly flat through the middle of the range, and null among veterans — the pattern supports Hypothesis 2: reciprocity is a newcomer phenomenon that fades with experience. We do not read the one-week-to-one-month dip as substantively meaningful, and we stop short of describing the decline as strictly graded.

For newcomers, both the arrival increment (β_4) and the waiting-period term (β_2) are positive and statistically significant ($\beta_4 = 0.068$, $\beta_2 = 0.050$, both $p < .001$), whereas among veterans both are near zero. The positive waiting-period term means that newcomers begin helping more even before their answer arrives. The arrival increment is measured net of that divergence, but a nonzero pre-trend in this stratum is itself a caution: anticipation,

exposure, or selection channels are active here, and the design cannot fully separate them from reciprocity (Section 6.5).

The tenure gradient is not an artifact of pooling structurally different entry cohorts. Because calendar year enters the design only as an exact-matching stratum, treated and control questions within a pair always come from the same year. To probe heterogeneity *across* years, we re-estimate Model A on the pre-2020 subsample — before the generative-AI disruptions of 2022–2023 — both pooled and among newcomers. The pooled pre-2020 arrival hazard ratio is 1.04 (versus 1.05 in the full sample), and the newcomer association is essentially unchanged (arrival HR = 1.07 pre-2020 versus 1.07 in the full sample; Appendix Table E.12). The tenure gradient was thus already present on the pre-generative-AI platform, not driven by recent cohorts.

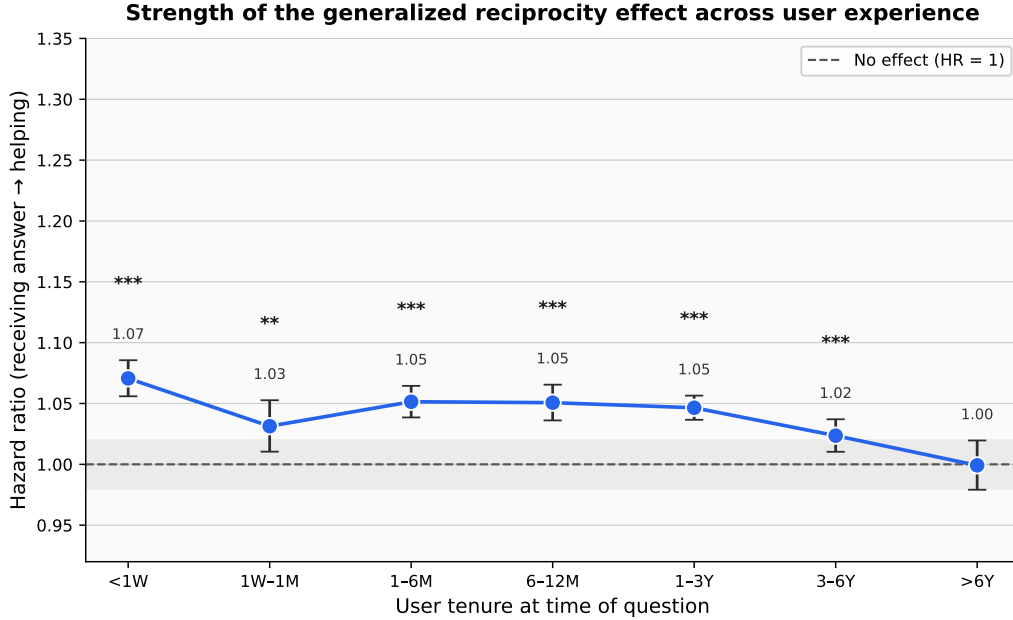


Figure 3: Strength of the generalized reciprocity effect across user experience. Each point shows the arrival-increment hazard ratio (β_4) estimated separately within a tenure bucket; error bars indicate 95% confidence intervals, and the dashed line marks no effect (HR = 1). Stars denote significance (** $p < .001$; ** $p < .01$).

Table 7: Cox Regression Results: Effect of Receiving an Answer on Helping Hazard

	< 1 Week	1 Wk– 1 Mo	1–6 Mo	6–12 Mo	1–3 Yr	3–6 Yr	> 6 Yrs
<i>Treatment effect: increment at answer arrival (β_4)</i>							
Answer arrival ($\times$ Post-Answer Received)	0.0683***	0.0309**	0.0502***	0.0494***	0.0455***	0.0233***	-0.0008
	(0.0071)	(0.0105)	(0.0063)	(0.0071)	(0.0048)	(0.0067)	(0.0103)
<i>Hazard Ratio</i> [95% CI]	[1.06, 1.09]	[1.01, 1.05]	[1.04, 1.06]	[1.04, 1.07]	[1.04, 1.06]	[1.01, 1.04]	[0.98, 1.02]
<i>Waiting period (β_2, parallel-trends diagnostic)</i>							
Received Answer $\times$ Post-Question	0.0504***	-0.0266*	-0.0147*	-0.0062	0.0000	-0.0107	-0.0167†
	(0.0070)	(0.0105)	(0.0063)	(0.0072)	(0.0049)	(0.0066)	(0.0101)
N	4,029,008	1,551,586	4,045,952	2,983,180	6,424,968	3,614,184	1,700,882
Events	368,780	238,189	694,001	537,263	1,037,470	428,036	132,234

Notes: N = unique questions (treated + control) in the Cox sample. Within each column, treated vs. control counts can differ because tenure is defined per question.

β_4 (the answer-arrival increment) is the difference-in-differences treatment effect; β_2 (waiting period) is reported as a parallel-trends diagnostic and is not added to the effect.

Cox-table standard errors are model-based; matched-pair bootstrap 95% CIs for the pooled arrival increment (β_4) are in Table D.11.

Events = helping events in the full analysis sample (after time rounding). When interval rows exceed the fit cap, estimation uses a matched-pair subsample, but Events still refer to the full sample.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.1$

5.5. Response Time Moderation (H3)

To test Hypothesis 3, which predicted that faster responses strengthen the reciprocity effect, we estimate the arrival increment and its response-time moderation separately by tenure bucket (Table 8); Table 9 and Figure 4 give a more flexible discrete-bin view. The evidence offers qualified support for the immediacy prediction. The continuous log-response-time interaction with the arrival increment (γ) is negative in every tenure bucket and statistically significant in all but the most senior: users who waited less for their answer responded more strongly, the direction H3 predicts. The moderation is strongest early in the trajectory (about -0.07 to -0.09 from the newcomer through one-year buckets), weakens with tenure (-0.05 at one to three years, -0.02 at three to six), and is indistinguishable from zero among users with more than six years. The discrete bins tell a consistent, if noisier, story. Users answered within 30 minutes barely respond at all, while those answered between 30 and 60 minutes respond most strongly of anyone (1.19 [$1.17, 1.21$]). It declines toward one over the following day (1.06 at one to two hours, 1.02 – 1.05 through one day), with a smaller secondary elevation at one to three days (1.09 [$1.07, 1.11$]). The pronounced early peak is closer to the “sweet spot” pattern originally hypothesized than to either a monotone speed gradient or a flat null; we return to the late elevation below.

Two qualifications temper this reading. First, the waiting-period gap carries a response-time moderation of its own (δ), positive in every bucket. This is the counterpart of the β_2 diagnostic: it reflects how the pre-answer gap itself varies with answer speed — an anticipatory or selection signal, not a response to receiving an answer. Because δ is not part of the arrival increment, the immediacy pattern we report is a property of that increment specifically, not of treated users’ overall activity trajectories. Second, we re-estimated each response-time bin with the same answer-quality controls (Appendix Table E.16). This adjustment is deliberately aggressive. Vote score and acceptance are strongly post-treatment; acceptance in particular requires the asker to return to the site, part of the very re-engagement pathway through which reciprocal helping materializes. Conditioning on them therefore absorbs some of the effect itself, not only the quality confound. The clearest sign of this over-adjustment is that under the controls the fastest bins fall significantly *below* one (0.95), which has no plausible causal reading. Even so, the 30–60-minute peak falls only from 1.19 to 1.03 (95% CI [$1.02, 1.04$], $p < .001$) and remains the significant early-window maximum, and the one-to-three-day elevation persists (1.09). We read the surviving peak as

genuine support for the immediacy pattern: answer timing retains explanatory content that observable answer quality does not absorb, even under an adjustment biased against finding it. We cannot, however, cleanly apportion the peak's absolute magnitude between timing and quality.

Table 8: Response Time Moderation of the Reciprocity Effect

	< 1 Week	1 Wk– 1 Mo	1–6 Mo	6–12 Mo	1–3 Yr	3–6 Yr	> 6 Yrs
<i>Treatment effect: arrival increment (β_4, at mean response time)</i>							
Answer arrival (net post-answer)	0.0101 [†]	0.0178*	0.0348***	0.0417***	0.0501***	0.0353***	0.0139
	(0.0057)	(0.0086)	(0.0052)	(0.0060)	(0.0041)	(0.0056)	(0.0086)
<i>Response-time moderation (γ = arrival; δ = waiting-period diagnostic)</i>							
Answer arrival $\times$ log(RT) (γ)	-0.0704***	-0.0778***	-0.0910***	-0.0863***	-0.0547***	-0.0228***	-0.0015
	(0.0048)	(0.0068)	(0.0039)	(0.0043)	(0.0028)	(0.0036)	(0.0055)
Waiting period $\times$ log(RT) (δ)	0.0236***	0.0725***	0.0892***	0.0829***	0.0670***	0.0453***	0.0278***
	(0.0039)	(0.0060)	(0.0035)	(0.0038)	(0.0024)	(0.0031)	(0.0045)
N	4,029,008	1,551,586	4,045,952	2,983,180	6,424,968	3,614,184	1,700,882
Events	368,780	238,189	694,001	537,263	1,037,470	428,036	132,234

Notes: γ and δ enter response time linearly in standardized log hours; the flexible discrete-bin counterpart of this specification is Table 9 and Figure 4.

Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; [†] $p < 0.1$

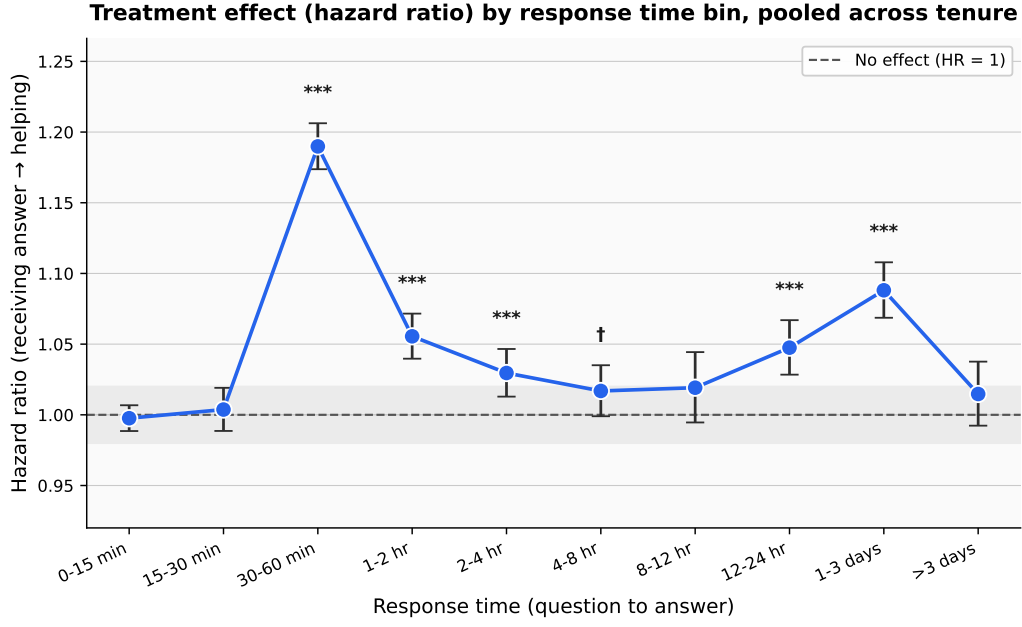


Figure 4: Treatment effect by response-time bin, pooled across all tenure buckets. Each point represents the arrival-increment hazard ratio (β_4) estimated within that response-time bin, relative to the matched control. Error bars indicate 95% confidence intervals; the dashed line marks no effect ($HR = 1$). Estimates correspond to Table 9; the continuous log-response-time specification is in Table 8.

Table 9: Treatment Effect by Response-Time Bin

Response time	HR	95% CI	N	Events
0-15 min	1.00	[0.99, 1.01]	17,412,740	2,484,407
15-30 min	1.00	[0.99, 1.02]	13,934,456	1,812,023
30-60 min	1.19***	[1.17, 1.21]	13,549,736	1,744,784
1-2 hr	1.06***	[1.04, 1.07]	13,211,144	1,698,505
2-4 hr	1.03***	[1.01, 1.05]	12,958,927	1,663,748
4-8 hr	1.02†	[1.00, 1.04]	12,762,008	1,639,963
8-12 hr	1.02	[0.99, 1.04]	12,464,162	1,601,705
12-24 hr	1.05***	[1.03, 1.07]	12,645,932	1,626,049
1-3 days	1.09***	[1.07, 1.11]	12,592,667	1,635,906
>3 days	1.01	[0.99, 1.04]	12,391,574	1,628,425

Notes: Each row fits Model A to treated questions in that response-time bin plus the full no-answer control pool. Bins are cut on the actual question-to-answer latency ($T_A - T_Q$); because the observation window closes two days after the answer, every treated question retains a full two-day post-answer phase, so β_4 is identified in all bins including >3 days. HR is the answer-arrival increment (β_4 , `is_treated_active`) at answer arrival.

Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.1$

Answer quality as an alternative explanation.. A natural concern is that response time stands in for answer quality. People who answer within minutes tend to post something terse or link-only, whereas those who take longer tend to write fuller, more detailed answers — so what prompts users to reciprocate may be the quality of the answer they got, not its timing. The bin-level re-estimation above is the direct test of this concern for the response-time pattern, and the 30–60-minute peak survives it. For transparency, Appendix Table E.14 reports how the pooled *level* of the arrival increment shifts as the same three quality controls are entered one at a time. Answer length — the least endogenous signal, fixed by the answerer at composition — barely moves it (from 1.04 to 1.03). The more endogenous vote-score and acceptance signals pull it further, to 1.01 and then below one (0.98); a score-coding sensitivity check accompanies these fits (Appendix Table E.17). As above, these level estimates are bounds, not estimates of the effect net of quality: conditioning on post-treatment signals partials out part of the effect itself. The genuine residual caveat is that our quality measures are coarse — character count does not distinguish worked code from boilerplate — so we cannot rule out that answer quality, imperfectly measured, remains a rival to a timing-based reading of the response-time pattern.

One interpretation of the response-time pattern draws on platform session dynamics: users who post a question may remain actively engaged while waiting, elevating helping during the waiting period regardless of whether an answer arrives. On this account, a fast answer might discharge that pending engagement rather than trigger graded reciprocal gratitude. This interpretation is speculative — we do not observe session boundaries (logins, logouts, inactivity gaps). The answer-quality account above is an equally plausible reading of the same pattern, and a third, purely selective account is that users whose questions attract answers were simply more active to begin with. The design cannot adjudicate among these, and we present the session-dynamics reading as one candidate among several rather than as an established mechanism.

Figure 5 provides descriptive context, comparing users who have received an answer (red) with users who will receive one but are still waiting (blue), alongside matched controls. In the first half-hour after the question — when only the fastest answers have arrived — users still waiting help at the highest rate, consistent with the mechanical elevation of users who have just asked a question and remain on the platform. From roughly an hour onward the ordering flips: users who have received an answer help at a modestly higher rate than those still waiting and than controls, consistent with a small post-answer elevation. Figure 6 disaggregates this comparison by tenure bucket and shows the same qualitative pattern in every stratum. These raw-rate orderings are descriptive: the differences are small, and answer-arrival timing is not random within the treated group, so we do not read them as independent evidence for any one mechanism — but they are consistent with the modest arrival increments the models estimate.

6. Discussion

6.1. Methodological Contribution

Researchers who study generalized reciprocity from platform data face a persistent difficulty: the people who seek help are the same people who give it. A correlation between help received and help given can therefore reflect nothing more than how active a user already was. Prior observational studies that report positive associations without accounting for this selection (Baker and Bulkley, 2014; Chen et al., 2018; Joyce and Kraut, 2006) may have overstated the role of reciprocity. The matched DiD survival design developed here addresses the confound: it compares changes in helping behavior, before

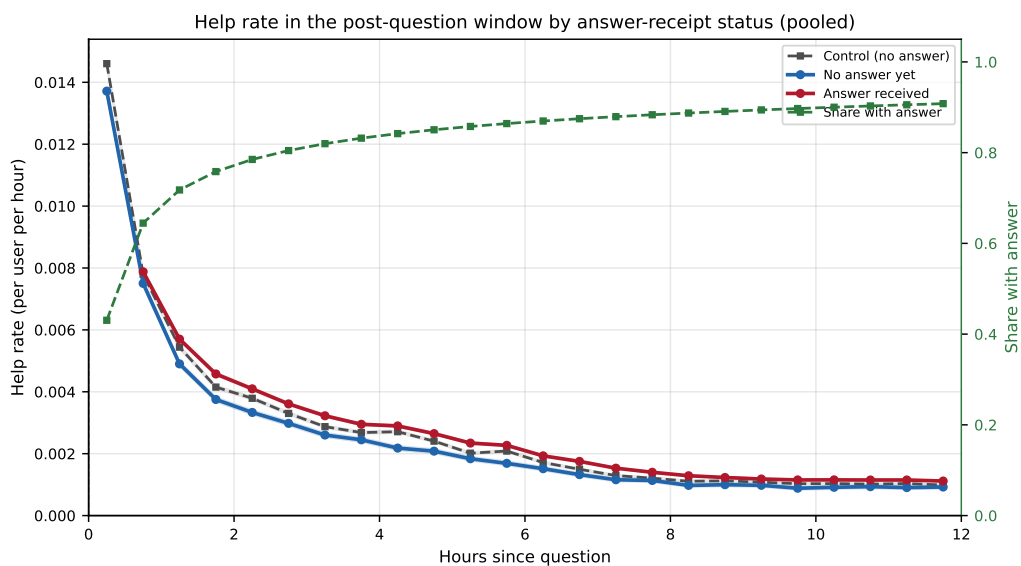


Figure 5: Help rate in the post-question window by answer-receipt status, pooled across all tenure buckets. Grey dashed: matched control users who received no answer. Blue: treated users who have not yet received an answer (No answer yet). Red: treated users who have already received an answer (Answer received). Green dashed (right axis): cumulative share of treated users who have received an answer by each hour.

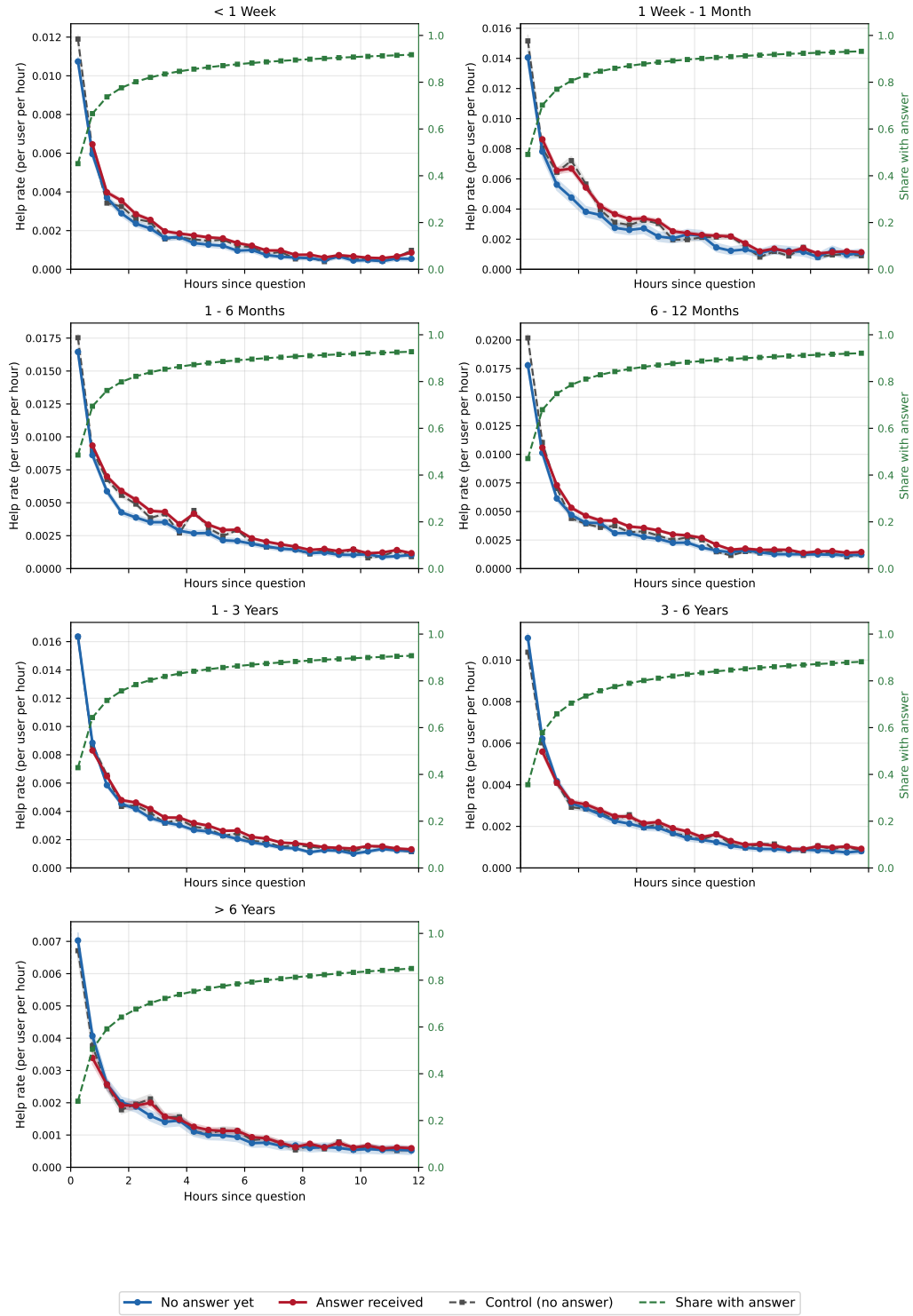


Figure 6: Help rate in the post-question window by answer-receipt status, separately by tenure bucket. Grey dashed: matched control users who received no answer. Blue: treated users who have not yet received an answer (No answer yet). Red: treated users who have already received an answer (Answer received). Green dashed (right axis): cumulative share of treated users who have received an answer by each hour.

and after question posting, between users whose questions received answers and propensity-score-matched users whose questions did not. Entering the waiting period as an explicit term makes any pre-answer divergence visible rather than silently absorbed into the treatment effect. In the present specification, the pooled contrast is small (arrival-increment $HR = 1.05$) and steeply newcomer-concentrated — substantially smaller than many earlier reports, consistent with activity confounding in those designs. Our estimate carries a caveat of its own: it is sensitive to controls for the quality of the answer received, so it bounds, rather than isolates, the association with answer receipt per se (Section 6.5).

6.2. Reciprocity as a Contributor-Recruitment Mechanism

Users tend to respond to being helped less the longer they have been on the platform: the newest respond most ($HR = 1.07$), and the most experienced not at all (arrival $HR = 1.00$). We cannot see why. Reciprocity, selection into being answered, the quality of the answers people received, or the displacement of general moral motives by community-specific incentives could each produce this gradient, and our design tracks what users did rather than what moved them. If reciprocity is what fades, then it plays a narrow role: users respond at the point of entry, before the community’s own incentives take hold.

As users gain experience, the community’s own incentives may crowd out the general reciprocal impulse — the norm-displacement account developed in Section 2.3. Experienced users on Stack Overflow have reasons to help that newcomers do not yet have: reputation, visible status, a stake in the community. That the most experienced users do not respond at all fits this account, though it fits selection into being answered equally well.

This reframing also helps account for the inconsistency of findings in the prior literature. Researchers who sample mostly experienced users, or who pool across experience levels dominated by experienced contributors, should find weaker reciprocity effects under this account. Those who catch users early should find stronger ones. Prior findings may vary, then, not because anyone erred, but because users themselves differ in how they respond.

6.3. The Response-Time Pattern and Its Interpretation

The response-time analysis tested the prediction, derived from laboratory studies of gratitude, that faster help produces stronger reciprocity. The pattern offers it qualified support. The continuous interaction runs in the

predicted direction — the arrival increment is larger for faster answers in every tenure bucket — and the discrete bins locate the effect in an early response-time window, with a pronounced peak for answers arriving between 30 and 60 minutes. This is broadly consistent with an emotional-immediacy account, in which help that arrives while the need is still salient elicits a stronger reciprocal response. One feature counsels caution at the outset: the very fastest answers (within 30 minutes) show no elevation, which a pure immediacy account does not predict.

Two non-gratitude readings could also generate an early-window peak. The first is answer quality, which the Results adjudicate directly (Section 5.5). The peak survives controls biased against it — genuine, if bounded, support for the immediacy reading — though its absolute magnitude cannot be cleanly apportioned between timing and quality. The second is platform session dynamics: an answer that arrives while a user is still in an active session may catch them at a moment of elevated engagement, whereas an answer that arrives after the session has ended cannot. We advance both readings as conjectures — our data record neither answer text nor session boundaries, so neither can be cleanly distinguished from a genuine immediacy effect or from selection. A secondary elevation at multi-day delays persists under the controls, but because it sits far from the immediacy window we cannot attach a clean mechanism to it and do not build on it.

The response-time pattern bears only weakly on the norm-displacement account. The arrival increment and its response-time moderation are both near zero among the most experienced users, which is at least consistent with the mechanism as a whole having faded rather than merely dampened. We do not lean on the response-time pattern as evidence for displacement.

6.4. Implications for Platform Design

Because the response we observe is modest and concentrated among newcomers, designers can use it mainly to recruit new contributors rather than to keep established members contributing — and even for recruitment, the gain is small in absolute terms.

Designers who want to grow their contributor base should therefore make sure that new users get answers to their questions. What they should not expect is a large return. Newcomers who received an answer went on to help roughly 0.65 percentage points more often per observation window than their matched counterparts ($HR = 1.07$; Table 7). Put differently, about 155

newcomer questions must be answered to produce one additional act of helping (Table 10). Designers gain about as much per answer further along the tenure range — roughly 144 questions per additional act of helping — because longer-tenured users already help more often, which offsets their weaker response. The case for concentrating on newcomers therefore rests not on a bigger yield per answer but on where users respond at all. The most experienced users do not respond (arrival HR = 1.00), so answering their questions buys a designer nothing. Designers thus have whatever leverage reciprocity offers at the point of entry — and not much of it anywhere.

Table 10: Absolute Effect Sizes and NNT

Row	Base risk	HR	ARD	NNT	NNT CI
< 1 Week	0.0915	1.071	0.00647	154.7	127.7–195.3
1 Week - 1 Month	0.1535	1.031	0.00481	207.9	123.7–624.9
1 - 6 Months	0.1715	1.051	0.00882	113.3	90.4–151.2
6 - 12 Months	0.1801	1.051	0.00913	109.6	84.8–153.8
1 - 3 Years	0.1615	1.047	0.00751	133.2	109.6–169.2
3 - 6 Years	0.1184	1.024	0.00279	358.0	227.9–819.3
> 6 Years	0.0777	0.999	-0.00007	—	—
AllData_Main	0.1411	1.049	0.00692	144.5	130.3–161.9

Notes: HR/ARD/NNT are based on the answer-arrival increment (β_4), the primary (headline) estimand.

We can offer designers only tentative support for acting on response speed. Users answered early responded most, and those answered between 30 and 60 minutes responded most of all (Tables 8 and 9), which is consistent with getting prompt answers to newcomers. But the effect is small, and we cannot fully separate how fast an answer arrived from how good it was, even though the 30–60-minute advantage does survive quality controls. We therefore stop short of recommending interventions built specifically on speed. Designers stand on firmer ground making sure that newcomers’ questions get answered at all, and that users notice when they do, than tuning for a particular response window.

Designers face a further complication as AI-generated answers become more common on question-and-answer platforms (see also the GAI-era limitation below). If people increasingly get their answers from an AI assistant rather than from another person, the human-to-human exchange that could sustain generalized reciprocity may thin out. Because the users who respond

at all are newcomers, that substitution could weaken exactly the onboarding channel through which new users now get their modest push into helping from other people.

6.5. Limitations

Several limitations constrain the conclusions that can be drawn from this analysis.

The most important identification caveat is right-truncation and churn: users who post a question and leave the platform before the fixed post-question window ends are under-observed for the outcome in ways correlated with both treatment and helping. The pipeline imposes no minimum-activity, minimum-tenure, or re-observation filter, so a user who asks a question, receives no answer, and leaves immediately is *retained* as a control and contributes zero help events over a fully observed window; 95.5% of control questions record no post-question helping (94.8–97.2% across tenure buckets). We do not implement a Heckman-type correction, because no matching covariate provides a credible exclusion restriction and the time-varying recurrent-event Cox model is not natively compatible with a selection equation. Matching does, however, restrict the estimand to a region of common support: of the roughly five million unanswered questions in the pre-matching sample, about half have no answered counterpart within the same calendar-year, tag, and tag-count cell and are dropped, so the retained control pool is drawn from more readily answered tags than the discarded tail. The reported contrast is therefore the effect for answered questions that have an on-support unanswered counterpart; to the extent that the discarded, less-answerable questions would have generated even less helping, the matched comparison is conservative rather than inflated for the churn-prone population this concern highlights. Discarding is not concentrated in the newcomer stratum that carries our central result — the < 1 week bucket is dropped at close to its arm-wide average rate.

Our helping outcome is the posting of an answer to another user’s question (answers to one’s own question do not count). The decomposition (Table 6) shows this choice is not what generates the effect: comments alone carry only a slight elevation, and adding them dilutes the estimate. Two further low-cost reciprocal channels that novices may favor nonetheless sit outside the outcome. Edits of others’ posts we observe and report separately (edits-only arrival HR 1.16), but keep out of the headline outcome because they index curation more than costly contribution; upvotes are absent from

our processed data and would require re-parsing the raw dump to recover. To the extent that newcomers reciprocate disproportionately through these channels, our answers-anchored estimates miss that portion of reciprocal behavior: the reported hazard ratios describe costly contribution specifically, not a complete accounting of reciprocity. We also cannot cleanly separate a reciprocal response to receiving help from selection into being answered.

Beyond identification, the design captures a behavioral association but cannot identify the psychological mechanism producing it. Gratitude, moral obligation, incentive displacement, and platform-session dynamics are all *proposed* interpretations consistent with parts of the pattern. The data do not let us distinguish among them, and we do not observe session boundaries, reputation salience, or emotional states. The most consequential competing explanation is answer quality, which we adjudicate in the Results (Section 5.5). The response-time peak survives controls biased against it, but the pooled *level* of the increment does not, so our estimates bound rather than isolate the association with answer receipt net of quality. Because those quality measures are coarse, we cannot rule out that answer quality, imperfectly measured, remains a rival to a reciprocity reading of the effect. Future work combining behavioral records with experience-sampling measures of emotional state or moral reasoning during platform use could help disentangle these pathways.

The identification strategy rests on the assumption that, conditional on the matched covariates and the DiD structure, answer receipt is approximately independent of the potential outcomes for helping behavior. The propensity score matching substantially improves the comparability of treated and control observations, as confirmed by post-matching balance diagnostics. Unobserved time-varying factors, including changes in workload, access to alternative information sources, or exogenous variation in a user’s availability, could nonetheless bias the estimates in ways that the design cannot rule out.

A specific unobserved channel of this kind is exposure. Posting a question raises the asker’s profile visibility. Other users who encounter the question may view the asker’s profile and contribution history, and this salience can generate reputation concern or social-obligation pressure — distinct from gratitude for a received answer, yet persisting into the post-answer window. The waiting-period covariate (*treatedPostQuestion*) absorbs the baseline effect of the act of asking, but not any residual visibility that continues after the answer arrives, so the design cannot fully separate gratitude for answer

receipt from this exposure channel. The threat is to mechanism attribution rather than to the existence of the behavioral pattern: the arrival increment remains a well-defined estimate of the behavioral change at answer receipt, but part of any elevation could reflect exposure rather than reciprocity. A natural probe would be a placebo test: do unanswered questions with *high* view counts show a helping lift in a pseudo-post-answer window? The archival data cannot support it. The Stack Overflow dump records *View-Count* only as a single cumulative snapshot at the dump date, so a question’s view count conflates age, topic popularity, and years of later (largely search-engine) traffic with the contemporaneous exposure such a placebo would require. Testing the exposure channel directly would require time-stamped view data or repeated dump snapshots that do not exist for our study period.

Response time may partially proxy for question characteristics not fully captured by the tag-level accepted-answer rate, since questions that attract fast responses may differ systematically from those that wait. Future work with random treatment or finer-grained measures of question difficulty, such as text-based complexity or specificity scores, could provide a cleaner test of the response-time mechanism.

The session-dynamics and re-engagement interpretations rest on indirect evidence: discrete-bin patterns, continuous speed interactions, and descriptive help-rate figures. Direct tests would require session-level data. We treat these accounts as conjectures, not established mechanisms.

Stack Overflow is a large, relatively anonymous, gamified platform built around technical question-answering. The patterns we estimate may not transfer to smaller or relation-based communities (enterprise Slack channels, academic peer review, intimate hobby forums) where social ties and non-status motives differ. We therefore refrain from claiming that generalized reciprocity “likely operates” elsewhere; at most, our estimates describe one platform’s pre-/early-generative-AI era.

The analysis window spans Stack Overflow’s inception through 1 April 2025 and should be read as a *historical snapshot*, not a prescriptive guide for contemporary platform design. Since the arrival of generative-AI tools in late 2022, Stack Overflow’s ecosystem has been structurally disrupted: active question volume and human-to-human answering have declined sharply. Our own data carry a trace of this decline: the per-window helping rate is markedly lower in the post-2020 portion of the sample, about 0.057 versus 0.164 helping events per matched window before 2020. The human-answer reciprocity associations we estimate were formed when community answer-

ing was the norm; their relevance to platforms increasingly mediated by AI-generated responses is uncertain. The pattern is not, however, confined to the recent AI-affected period: restricting the sample to the pre-2020 era leaves both the pooled and newcomer associations essentially unchanged (Appendix Table E.12). Because that pre-2020 subsample supplies most of the analysis — 79% of matched rows and 91% of helping events — this is a compositional check rather than an independent replication on disjoint data; we read it as evidence that our estimates characterize the pre-generative-AI platform, not as a second, independent confirmation of the effect.

Finally, the analysis establishes that any short-run association declines with experience but does not characterize what replaces it. The theoretical argument implicates community-specific norms and status motives, but testing this account would require integrating behavioral analysis with direct measures of those alternative mechanisms, for instance, through surveys or experimental manipulations of reputation visibility.

7. Conclusion

Generalized reciprocity is widely theorized as a mechanism sustaining cooperation on knowledge-sharing platforms, but observational evidence runs into a persistent confound: the users who ask for help are the users who give it, whether or not their own questions get resolved. Our matched difference-in-differences survival design confronts that confound directly. Users who received an answer went on to help slightly more often than their matched counterparts (arrival increment $HR = 1.05$); newcomers account for nearly all of it (arrival $HR = 1.07$), and veterans for none. Newcomers respond in the pre-2020 data, before generative AI, much as they do in the full sample (Appendix Table E.12), yet even their response is modest in absolute terms. Newcomers appear to help others a little more after someone has helped them. But our estimates cannot, on their own, separate that story from selection into being answered or from the quality of the help that arrives; distinguishing them remains a task for designs with exogenous variation in whether, and how quickly, help arrives. What the pattern does support is a reframing: on this platform, generalized reciprocity works less as a general engine of cooperation among established members than as a contributor-recruitment channel at the point of entry — a narrow window, before community-specific incentives take hold, in which receiving help and beginning to give it go together. For a designer, that window is the moment worth protecting.

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Ethics statement

This study analyzed de-identified, publicly available archival records from Stack Overflow. No interaction with human participants occurred, and no personally identifiable information was collected or analyzed. Ethical approval was not required under applicable institutional policies for secondary analysis of public platform data.

Data availability

Stack Overflow public data dumps were obtained from the Internet Archive (<https://archive.org/details/stackexchange>). Analysis code and step-by-step instructions to reproduce all tables and figures are available in a public repository (URL withheld for blind review). Processed analytical datasets used in the matching and survival analyses are available from the corresponding author upon reasonable request.

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Appendix A. Replication materials

Analysis code and instructions to reproduce all tables and figures are available in a public repository (URL withheld for blind review).

Appendix B. Matching Diagnostics

Figure B.7 shows sufficient common support between the treatment and control groups: the propensity-score distributions of the original treated and control groups are similar, and the distribution of the matched control group is nearly identical to that of the treated group.

Figure B.8 displays the standardized mean differences (SMDs) for the matching covariates before and after matching. Several covariates show noticeable imbalance before matching; after matching, all SMDs are close to zero.

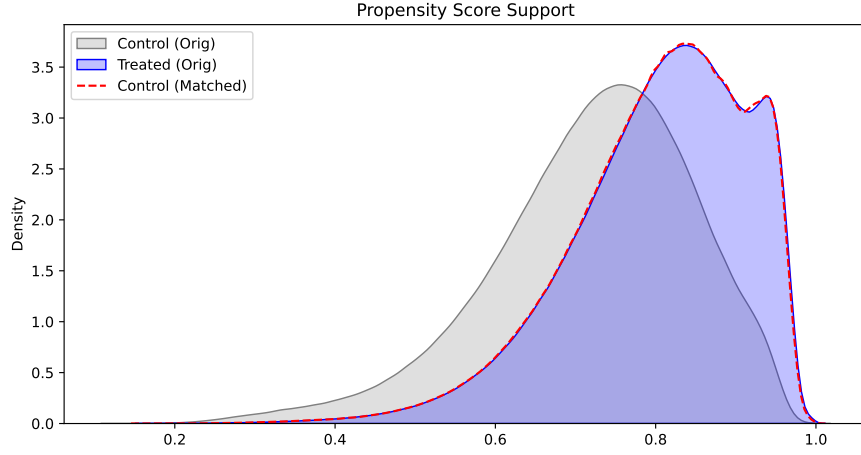


Figure B.7: Distribution of propensity scores of treatment group, control group, and matched control group

Appendix C. Help Rate Over the Observation Window by Tenure

Figure C.9 shows that treated and control help rates track each other closely before the question in every tenure bucket, supporting the parallel-trends logic within each stratum. The sharp spike at question time is common to both groups and is somewhat larger for controls; the small post-answer elevations estimated in Table 7 and Figure 3 (hazard ratios of a few percent) are below what these raw normalized rates can visually resolve.

Appendix D. Bootstrap Inference and Absolute Effects

Table D.11 reports matched-pair bootstrap 95% confidence intervals for the pooled Model A treatment effect — the difference-in-differences increment at answer arrival (β_4) — based on 100 replicates. Absolute effect sizes (absolute risk differences and numbers-needed-to-treat) are reported in the main text (Table 10, “Implications for Platform Design”).

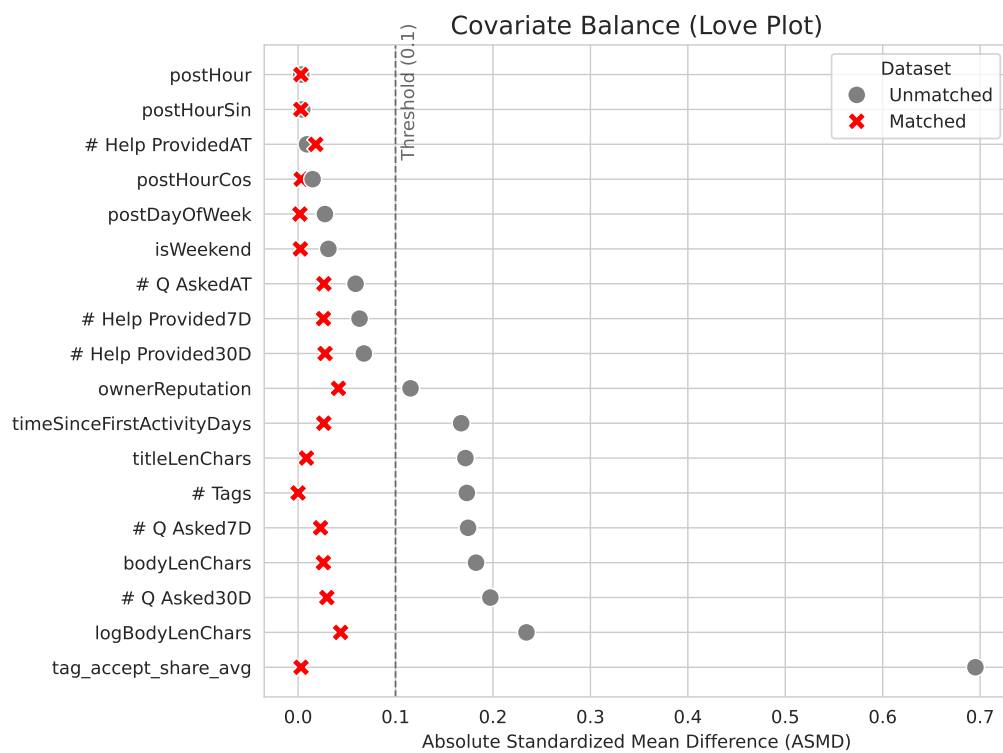


Figure B.8: Visualization of balance in the unmatched and matched dataset

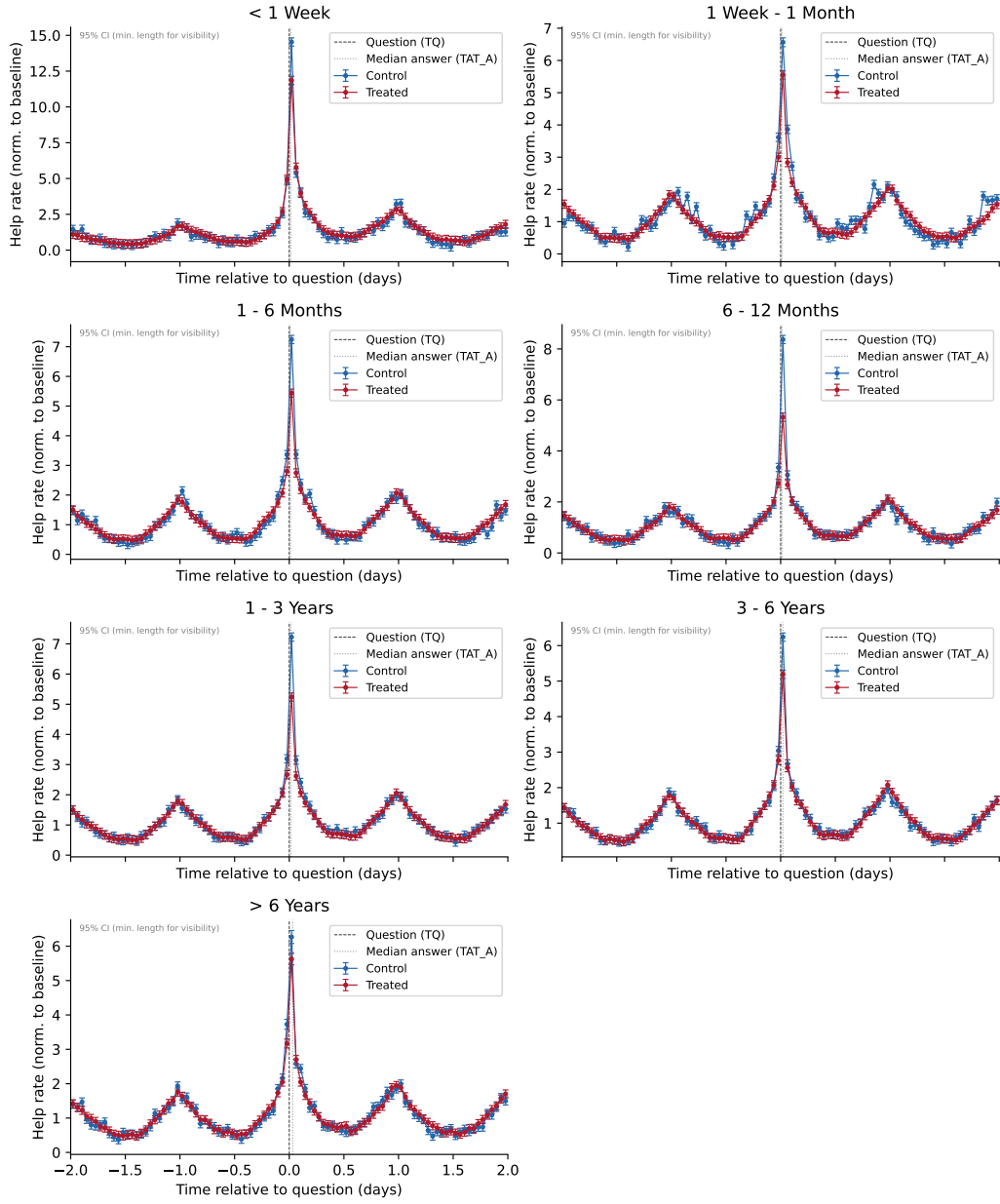


Figure C.9: Help Rate Over the Observation Window, by Tenure Bucket. Each panel plots the normalized help rate (relative to each group's pre-question baseline) for treated users (red) and matched control users (blue) over the ± 2 -day window around the question-posting time (T_Q , dashed vertical line). The dotted vertical line marks the median answer arrival time (T_A). Error bars indicate 95% confidence intervals.

Table D.11: Matched-Pair Bootstrap Uncertainty for the Answer-Arrival DiD Increment (β_4 ; Model A)

Scope	HR	Bootstrap 95% CI	Replicates	N
all	1.05	[1.03, 1.07]	100/100	24,349,760

Notes: Replicates resample whole matched pairs with replacement. HR is the answer-arrival increment $\exp(\beta_4)$, the difference-in-differences treatment effect.

Appendix E. Robustness: Cohort Heterogeneity, Observable Selection, and Answer Quality

Table E.12 addresses the concern that pooling entry cohorts from 2008 through 2025 conflates structurally different eras of the platform (reputation inflation, tag evolution, and the post-2022 generative-AI disruption). Because calendar year enters the matching design only as an exact stratum, treated and control questions within a pair always share a year; to probe heterogeneity *across* years we re-estimate Model A on the pre-2020 subsample, both pooled and in the newcomer (< 1 week) stratum. The pooled pre-2020 arrival hazard ratio is 1.04 (versus 1.05 in the full sample), and the newcomer association is essentially unchanged (arrival HR = 1.07 pre-2020 versus 1.07 in the full sample), indicating that the tenure gradient was already present before the generative-AI era. Because calendar year enters as an exact-matching stratum, the pre-2020 subsample is not disjoint from the full sample but a large sub-population of it, retaining 79% of matched rows and 91% of helping events (Table E.12). This check is therefore compositional — it confirms the association does not depend on the post-2020 cohorts — rather than an independent replication on separate data.

Table E.12: Cohort Heterogeneity Robustness

Specification	Arrival HR	95% CI	Waiting β_2	N	Events
ModelA_AllData	1.049	[1.044, 1.054]	-0.007*	24,349,760	3,435,973
ModelA_Newcomer	1.071	[1.056, 1.086]	+0.050***	4,029,008	368,780
ModelA_Pre2020	1.037	[1.031, 1.042]	-0.000	19,187,320	3,144,155
ModelA_Pre2020_Newcomer	1.068	[1.052, 1.085]	+0.071***	3,155,176	336,966

Notes: Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

Table E.13 reports the pooled Model A after adding pre-treatment observable selection controls (posting hour and weekday, number of tags, body and

title length, and asker reputation). View count is *not* included: it accrues after posting and is higher for answered questions, so conditioning on it would open a post-treatment / collider channel already excluded from the propensity score. The pooled arrival increment remains essentially unchanged once these pre-treatment observables are held fixed (arrival HR = 1.04, versus 1.05 in the baseline specification).

Table E.13: Observable Selection Controls

Specification	Arrival HR	95% CI	Waiting β_2	N	Events
baseline	1.049	[1.044, 1.054]	-0.007*	24,349,760	3,435,973
observable_controls	1.040	[1.036, 1.044]	+0.002	24,349,760	3,435,973

Notes: Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

Table E.14 reports the pooled speed-specification arrival increment as the three answer-quality controls (first-answer length, vote score, and acceptance) are entered one at a time. The gradient is informative about what each signal absorbs: answer length — the least endogenous signal, fixed by the answerer at composition — barely moves the estimate (1.038 baseline to 1.032); vote score brings it to 1.010; and the full specification, adding acceptance, pushes it below one (0.982), with the same pattern in the newcomer stratum (0.95; Table E.15). Because vote score and, especially, acceptance are post-treatment and endogenous to the outcome (accepting requires the asker to return to the site, the same re-engagement that produces helping), these level estimates are a bounding exercise rather than a causal decomposition: the association is robust to the terse-versus-detailed concern (length), and it is reversed only by the signals a bad-control logic predicts would over-adjust.

Table E.14: Answer-Quality Robustness

Specification	Arrival HR	95% CI	Waiting β_2	N	Events
speed_baseline	1.038	[1.034, 1.042]	+0.014***	24,349,760	3,435,973
speed_length_only	1.032	[1.027, 1.036]	+0.006**	24,349,760	3,435,973
speed_score_only	1.010	[1.006, 1.015]	-0.015***	24,349,760	3,435,973
speed_quality_controls	0.982	[0.978, 0.986]	-0.046***	24,349,760	3,435,973

Notes: Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

Table E.15 reports the newcomer (< 1 week) bucket under several specifications. The baseline arrival increment is 1.07. Adding the same pre-

treatment observable controls — still *without* view count — raises the arrival increment to 1.13, which indicates the newcomer association is not an artifact of *observable* selection. This gain warrants caution rather than reassurance, however: the waiting-period diagnostic rises in step, from $\beta_2 = +0.050$ to $+0.122$, essentially the magnitude of the arrival log-increment itself ($\log 1.13 = 0.121$). Under these controls, treated newcomers already diverge upward from their matched controls by nearly the full effect size *before* the answer arrives, so the two terms inflate together; we therefore read the specification as ruling out observable selection as the sole driver, not as independent corroboration of the arrival increment, and it sharpens the parallel-trends caveat that already attaches to this stratum. Adding answer-quality controls, by contrast, pushes the newcomer increment below one (arrival 0.95), the same pattern seen in the pooled effect and reinforcing that answer quality — not observable selection — is the most consequential confound the design does not resolve.

Table E.15: Newcomer Bucket Robustness (< 1 Week)

Specification	Arrival HR	95% CI	Waiting β_2	N	Events
ModelA_Newcomer_Baseline (< 1 Week)	1.071	[1.056, 1.086]	+0.050***	4,029,008	368,780
ModelA_Newcomer_Observable (< 1 Week)	1.129	[1.117, 1.142]	+0.122***	4,029,008	368,780
ModelB_Newcomer_Baseline (< 1 Week)	1.010	[0.999, 1.021]	+0.008	4,029,008	368,780
ModelB_Newcomer_LengthOnly (< 1 Week)	1.003	[0.992, 1.014]	-0.001	4,029,008	368,780
ModelB_Newcomer_ScoreOnly (< 1 Week)	0.979	[0.968, 0.990]	-0.022***	4,029,008	368,780
ModelB_Newcomer_Quality (< 1 Week)	0.950	[0.939, 0.960]	-0.052***	4,029,008	368,780

Notes: Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

Table E.16 revisits the discrete response-time bins after adding the same three answer-quality controls used in the pooled speed specification (acceptance, first-answer score, and length). Because each row already conditions on a response-time window, the quality-controlled fits use Model A covariates plus those three signals rather than continuous log-response-time interactions. Comparing columns shows how much of the bin-wise arrival pattern survives once observable quality is held fixed.

Table E.16: Treatment Effect by Response-Time Bin, With and Without Answer-Quality Controls

Response time	Baseline HR	95% CI	Quality HR	95% CI
0-15 min	1.00	[0.99, 1.01]	0.95***	[0.95, 0.96]
15-30 min	1.00	[0.99, 1.02]	0.95***	[0.94, 0.96]
30-60 min	1.19***	[1.17, 1.21]	1.03***	[1.02, 1.04]
1-2 hr	1.06***	[1.04, 1.07]	0.96***	[0.95, 0.98]
2-4 hr	1.03***	[1.01, 1.05]	0.96***	[0.95, 0.97]
4-8 hr	1.02†	[1.00, 1.04]	0.97***	[0.95, 0.98]
8-12 hr	1.02	[0.99, 1.04]	0.97**	[0.95, 0.99]
12-24 hr	1.05***	[1.03, 1.07]	1.01	[0.99, 1.02]
1-3 days	1.09***	[1.07, 1.11]	1.09***	[1.07, 1.11]
>3 days	1.01	[0.99, 1.04]	1.04***	[1.02, 1.06]

Notes: Each row fits Model A to treated questions in that response-time bin plus the full no-answer control pool. Bins are cut on the actual question-to-answer latency ($T_A - T_Q$); because the observation window closes two days after the answer, every treated question retains a full two-day post-answer phase, so β_4 is identified in all bins including >3 days. HR is the answer-arrival increment (β_4). Quality columns add acceptance, first-answer score, and length.

Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.1$

Table E.17 probes whether the answer-quality result depends on how the vote-score control is coded. The specifications in Table E.14 enter the score linearly (winsorized at the 5th and 95th percentiles and standardized); here we re-estimate the score-only and full-quality specifications, pooled and in the newcomer stratum, under two alternatives — a signed-log (asinh) coding and ordinal bins (1–2, 3–9, and ≥ 10 votes, reference ≤ 0) — at full data. We use asinh(score) rather than $\log(1 + \text{score})$ because roughly two million answers carry net-negative (downvoted) scores, for which $\log(1 + \text{score})$ is undefined. The below-one full-specification result is robust to the coding: the pooled arrival increment is 0.98 under all three codings and the newcomer counterpart 0.95, and if anything the better-specified signed-log and binned codings pull the estimate marginally *further* below one rather than back toward it. The one estimate that does not survive re-coding is the intermediate score-only step, which is significantly above one under the linear coding (1.01) but statistically indistinguishable from one under the signed-log and binned codings; we therefore do not read it as evidence of an above-one effect.

Table E.17: Answer-Score Coding Sensitivity

Specification	Score coding	Arrival HR	95% CI	Waiting β_2	N	Events
<i>Pooled</i>						
Speed baseline	—	1.038***	[1.034, 1.042]	+0.014***	24,349,760	3,435,973
+ Vote score	Linear	1.010***	[1.006, 1.015]	-0.015***	24,349,760	3,435,973
	Asinh	1.001	[0.997, 1.005]	-0.026***	24,349,760	3,435,973
	Ordinal bins	1.003	[0.999, 1.007]	-0.023***	24,349,760	3,435,973
+ Full quality controls	Linear	0.982***	[0.978, 0.986]	-0.046***	24,349,760	3,435,973
	Asinh	0.977***	[0.973, 0.982]	-0.052***	24,349,760	3,435,973
	Ordinal bins	0.979***	[0.975, 0.983]	-0.050***	24,349,760	3,435,973
<i>Newcomer (< 1 Week)</i>						
Speed baseline	—	1.010†	[0.999, 1.021]	+0.008	4,029,008	368,780
+ Vote score	Linear	0.979***	[0.968, 0.990]	-0.022***	4,029,008	368,780
	Asinh	0.974***	[0.963, 0.985]	-0.028***	4,029,008	368,780
	Ordinal bins	0.976***	[0.965, 0.987]	-0.026***	4,029,008	368,780
+ Full quality controls	Linear	0.950***	[0.939, 0.960]	-0.052***	4,029,008	368,780
	Asinh	0.947***	[0.936, 0.957]	-0.056***	4,029,008	368,780
	Ordinal bins	0.949***	[0.938, 0.959]	-0.054***	4,029,008	368,780

Notes: Codings of the answer’s vote score, all re-expressing the same control: *Linear* is the control tabled in Table E.14 (raw *Score* winsorized at the 5th/95th percentiles and standardized); *Asinh* is the inverse hyperbolic sine $\text{asinh}(\text{score}) = \ln(\text{score} + \sqrt{\text{score}^2 + 1})$ (a signed log, concave in the score and defined for the negative scores present), likewise winsorized and standardized; *Ordinal bins* are indicators for 1–2, 3–9, and ≥ 10 votes (reference score ≤ 0). Arrival-HR stars test H_0 : HR = 1. The bin reference is score ≤ 0 and the concave coding uses asinh rather than $\log(1 + \text{score})$ because roughly two million answers carry net-negative (downvoted) scores, for which $\log(1 + \text{score})$ is undefined. Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.1$

Appendix F. Implementation Details

All models are estimated using the `lifelines CoxTimeVaryingFitter` with a small ridge (L2) penalizer applied for numerical stability. The penalizer strength is $\lambda = 5 \times 10^{-3}$ for main-effect models and $\lambda = 10^{-2}$ for models that include response-time interaction terms (where the interaction covariates are correlated with the base treatment indicator). These values are sufficiently small that they introduce negligible shrinkage relative to the scale of the estimated effects, but they ensure stable Hessian inversion in large risk sets. All covariates are mean-centered before fitting; response-time interaction terms are additionally clipped to their 5th–95th percentile range and z-scored as described above. Cox-table standard errors are model-based (inverse-Hessian) and are *not* cluster-robust; dependence from matched pairs, reused controls, and repeated intervals can make those intervals too narrow.

For the pooled Model A DiD we therefore also report a matched-pair bootstrap 95% CI (100 replicates resampling whole `match_id` pairs) in Table D.11. For the pooled models (H1, spanning all experience levels), point estimation uses a random subsample of approximately 8 million user-interval rows drawn from matched pairs (preserving pair integrity) when the full interval frame exceeds the fit cap; table *Events* counts still refer to the full analysis sample. Tenure-bucket-specific models (H2, H3) use all available intervals within each stratum.